Trust but Verify: Diagnosing Causal Prior-Data Fitted Networks and Testing When to Defer

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Abstract

Causal prior-data fitted networks (PFNs) estimate conditional average treatment effects in one forward pass, but nothing in their output tells a user whether the data fit the prior and satisfy the identifying assumptions that their guarantees rest on. We diagnose the three released causal PFNs on a 50-cell simulation platform and four benchmarks. A generic TabPFN inside a doubly robust learner (TabPFN-DR) is more accurate than all three on most designs, so causal pretraining should be judged against it. CausalPFN's uncertainty is aleatoric: its 90% CATE intervals cover 1.00 in every cell, while its 95% ATE intervals cover 0.55 in the base design. No native signal flags violated overlap, CausalPFN's wrapper has two silent limits, and a confounding bias in CausalPFN and CausalFM does not vanish with n. We also propose certify-or-defer, a wrapper that needs no retraining: in-context cross-fitting yields a paired doubly robust estimate of the PFN's excess CATE risk over classical references, and a region-wise one-sided Holm test replaces the PFN only where it is significantly worse. A certificate states only that this test finds no excess risk relative to the references, not that the PFN is accurate. Under a protocol frozen before the method was built, a confirmatory test on four benchmarks and six held-out mechanisms gives a mixed verdict. Certify-or-defer is non-inferior to the PFN on all four benchmarks and significantly more accurate on ACIC-2016 and the held-out suite, but its detection AUROC on ACIC-2016 misses the pre-registered 0.80 (0.783 and 0.788), and 18.9% of its deferrals for CausalPFN do harm, above the 10% bound. TabPFN-DR remains more accurate than the wrapper on most benchmarks; post-verdict analyses trace most of the harm to a bias of the doubly robust score where overlap fails. A run pre-registered on 160 new ACIC-2017 tasks confirms the R-loss variant we recommend, though there the registered test did no harm either.

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1. Introduction

Causal prior-data fitted networks estimate treatment effects in a single forward pass. CausalPFN, Do-PFN and CausalFM are trained on large collections of simulated datasets and, given an observational table, return conditional average treatment effects (CATEs) without per-dataset tuning, CausalPFN and CausalFM also with intervals (Balazadeh et al., 2025; Robertson et al., 2025; Ma et al., 2026). Every guarantee they carry, however, is conditional: the posterior they approximate is the posterior under their training prior, and the CATE they target is identified from (X, T, Y) only under ignorability and overlap. Nothing in the output tells the user whether their data satisfy these conditions.

We ask three questions. **Q1:** when, and why, do causal PFNs fail on data they were not trained on? **Q2:** can such failures be detected from the observed data (X, T, Y) alone? **Q3:** can detection be turned into a decision that does no harm, so that a user keeps the PFN when it is good and replaces it when it is not?

We answer them under a protocol frozen before the method was built: the diagnostic hypotheses were registered before their runs, the success criterion before the pilots, and the final benchmarks and a held-out suite were run once, at the end. We report every outcome, including the failed criteria.

Contributions.

- **A diagnosis of the released causal PFNs** on a 50-cell development platform and four benchmarks (Section 4). Their accuracy relative to well-configured classical estimators depends on how well the data fit the prior, and a generic TabPFN inside a cross-fitted DR-learner is more accurate than all of them on most of our designs, which makes it the baseline that causal

pretraining has to beat. CausalPFN's uncertainty is aleatoric: its CATE intervals cover 1.00 in every cell, while its ATE intervals cover 0.55 in the base design and 0.00 under hidden confounding. No native signal flags violated overlap, and under hidden confounding the signals move the wrong way. We also identify two silent wrapper limits (an unsupervised SVD for $d > 99$ and a context cap) and a confounding bias that does not vanish with n.

- **An evaluation audit.** The published IHDP ATE errors of CausalPFN are scored against a constant truth, which inflates them $8.7\times$; the RealCause LaLonde "PEHE" is against noisy ITEs; and the benchmark choice decides which method looks best. Where their authors report errors we can recompute (10 ACIC-2016 datasets in their normalized units, and Jobs), our use of CausalFM and Do-PFN reproduces them (Section 4.5). We release the harness, the frozen protocol and the held-out suite.
- **Certify-or-defer** (Section 5), a model-agnostic wrapper that needs no retraining: in-context cross-fitting, a paired doubly robust estimate of the PFN's excess CATE risk over classical references, and a region-wise deferral rule with Holm control. Under standard product-rate conditions the deferral has asymptotic size α and power 1 and does no harm in expectation beyond α times the worst loss (Proposition 5.2). A certificate is a one-sided test of no excess risk relative to the references, not evidence of accuracy.
- **A pre-registered evaluation** (Section 6): two pilots on development data, with the second pilot's detection criterion failed for CausalPFN and passed for CausalFM, and a confirmatory test on IHDP, the full ACIC-2016, both LaLonde samples and a held-out suite of six mechanisms. The wrapper is non-inferior to the PFN everywhere and passes the do-no-harm claim for CausalFM, but it fails detection on ACIC-2016 for both PFNs and the harm bound for CausalPFN, so the generality claim fails (Table 1). Post-verdict analyses trace the harm to a bias of the doubly robust score where overlap fails. A run pre-registered on 160 new ACIC-2017 tasks confirms the R-loss variant we recommend, but the registered test did no harm there either (Appendix I).

Figure 1 previews the central diagnostic fact: the variance a causal PFN reports is, to first order, outcome noise.

2. Related Work

Causal prior-data fitted networks. PFNs are transformers trained on datasets simulated from a prior so that a forward pass approximates the posterior predictive distribution (Müller et al., 2022; Nagler, 2023); TabPFN made this practical for tabular prediction (Hollmann et al., 2023; 2025). CausalPFN (Balazadeh et al., 2025) and CausalFM (Ma et al., 2026) extend the idea to treatment effects under backdoor adjustment (CausalFM also to front-door and instrumental settings); Do-PFN (Robertson et al., 2025) predicts interventional outcomes without a known graph, from a prior that also contains non-identified models. Follow-ups add partial graphs, continuous treatments, panel data, proximal inference, mixed experimental regimes and prior selection (Reuter et al., 2026; Stith et al., 2026; Talupula & Sharma, 2026; Muller et al., 2026; Gao et al., 2026; Zhou, 2026). All of them assume that the deployment data lie in the support of the prior and satisfy the identifying assumptions. The closest check, the judgment head of TCPFN (Talupula & Sharma, 2026), is supervised on labels from its own prior and is specific to panel data.

Corrections and uncertainty for PFN effect estimates. Melnychuk et al. (2026) show that PFN priors induce an ATE confounding bias that does not vanish with n, and correct it with a one-step update on martingale-posterior nuisances (MP-OSPC); FSP builds the correction into the pretraining labels (Zhang, 2026). Both target the ATE and leave CATE-level reliability open. Martingale posteriors (Fong et al., 2023) give PFN functionals uncertainty by predictive resampling (Nagler & Rügamer, 2025; Ng et al., 2025). An independent study reports 54% and 68% coverage of nominal 95% CausalPFN ATE intervals in two clinical scenarios at $n = 1000$ (Mourao et al., 2026).

Misspecification checks. In simulation-based inference, unreliable amortized posteriors are flagged by summary-space distances (Schmitt et al., 2021), calibration diagnostics (Hermans et al., 2022; Linhart et al., 2023) or robust posteriors (Ward et al., 2022); prior–data conflict and held-out predictive checks do the same for Bayesian models (Nott et al., 2020; Moran et al., 2024; Li & Huggins, 2026). These test the fit of the observed data, while the failures that matter for treatment effects are counterfactual.

CATE validation and hidden confounding. Doubly robust and orthogonal losses make CATE risk estimable from observational data (Robins et al., 1994; Chernozhukov et al., 2018; Foster & Syrgkanis, 2023; Kennedy, 2023; Nie & Wager, 2021). They underlie calibration (Xu & Yadlowsky, 2022; van der Laan et al., 2023), best linear projections (Chernozhukov et al., 2017), influence-function validation of causal models (Alaa & van der Schaar, 2019) and model selection, where no criterion is best across settings, a criterion tends to favour estimators built like itself, and Schuler et al. (2018) recommend the R-loss (Saito & Yasui, 2020; Mahajan et al., 2024; Curth & van der Schaar, 2023). Comparing two estimators by the difference of their doubly robust losses, in which τ^2 cancels, is standard there; we add

a size-controlled, region-wise deferral rule and in-context cross-fitting of a frozen foundation model. Our deferral is a per-dataset choice between fixed estimators made by a test, not the learned per-instance rejector of learning to defer (Madras et al., 2018; Mozannar & Sontag, 2020). Overlap can be characterized from (X, T) alone (Oberst et al., 2020; D'Amour et al., 2021), and falsification tests reject observational estimators with randomized data (Hussain et al., 2022). Unobserved confounding cannot be detected from $P(X, T, Y)$ in general; the response is sensitivity analysis (Kallus et al., 2019; Jin et al., 2023; Dorn & Guo, 2023; Oprescu et al., 2023; Chernozhukov et al., 2026), recently amortized with PFNs (Javurek et al., 2026). We did not find, in the literature we surveyed (through 2026-10-07), such a test between a frozen foundation-model estimator and classical references combined with a size-controlled deferral rule.

3. Setup and Protocol

Estimands. We observe $\mathcal{D} = \{(X_i, T_i, Y_i)\}_{i=1}^n$ i.i.d. from P, with covariates $X \in \mathbb{R}^d$ and a binary treatment T. Let $e(x) = \mathbb{P}(T = 1 \mid X = x)$, $\mu_t(x) = \mathbb{E}[Y \mid X = x, T = t]$ and $\tau = \mu_1 - \mu_0$, which is the CATE under ignorability and overlap (Rosenbaum & Rubin, 1983). A causal PFN f is a fixed, pretrained map from a dataset and a query point to an effect estimate, $\hat{\tau}_f(x) = f(\mathcal{D}, x)$, usually with an interval. We score point estimates on held-out test units by $\sqrt{\text{PEHE}} = (\frac{1}{m}\sum_j(\hat{\tau}(x_j) - \tau(x_j))^2)^{1/2}$, the ATE (mean CATE over the n training units) by its absolute error, and intervals by coverage and width.

Methods. We evaluate the three released causal PFNs for binary treatments: CausalPFN (package defaults, raw and with its temperature calibration), Do-PFN v1 and CausalFM (standard CATE model) (Balazadeh et al., 2025; Robertson et al., 2025; Ma et al., 2026). The classical baselines are a DR-learner with tuned LightGBM nuisances and 5-fold cross-fitting (Kennedy, 2023; Ke et al., 2017), a causal forest (Wager & Athey, 2018; Athey et al., 2019; Battocchi et al., 2019) and BCF (Hahn et al., 2020; Herren et al., 2025). We also run TabPFN v2 (Hollmann et al., 2025) as a T-learner (TabPFN-T; Künzel et al., 2019) and inside the same cross-fitted DR-learner (TabPFN-DR), and MP-OSPC with the authors' code (Melnychuk et al., 2026). All methods run in one harness on the same tasks and seeds; every baseline with nuisance models gets the same tuning grid; the PFNs are frozen, by design (Appendix A).

Data. *Development* uses a synthetic platform: a base design ($n = 1000$ training and 500 test units, $d = 10$ with 5 active covariates, additive outcome, linear effect, ATE and effect s.d. 0.5, propensity logit scale $\kappa = 1$) and one-factor-at-a-time sweeps along 15 axes (functional forms, heterogeneity, effect size, noise, covariates, scale, d, n, overlap, confounding, hard overlap violation, hidden confounding): 50 cells with 20 seeds each (Appendix B). The *final* benchmarks are IHDP-100 (B1; 672 training units, $d = 25$; Hill, 2011; Shalit et al., 2017), the full ACIC-2016 competition, 77 settings $\times$ 10 replications (B2; 4321 training units, $d = 58$; Dorie et al., 2019), and the Real-Cause LaLonde CPS and PSID samples (B3 / B4; 100 each; LaLonde, 1986; Dehejia & Wahba, 1999; Neal et al., 2020), whose released files contain only noisy potential outcomes, so "PEHE" there is an ITE-RMSE with a noise floor. The *held-out mechanism suite* H has six generators that lie outside the platform's axes and, in at least one respect, outside the published priors (Appendix B; misspecified nuisances, a step-function CATE, heavy tails, a high-cardinality categorical, an M-bias collider, instruments with hidden confounding), each at $n \in \{500, 2000\}$ with 20 seeds. H was written, and its code frozen, before any method was designed.

Protocol. The evaluation protocol was drafted before any method design and frozen, together with the method's success criterion, before the method was implemented. H was used for nothing else and the final evaluation was run once. The diagnosis phase did run the published protocols on B1, B3, B4 and a 10-setting subset of B2 to verify that our harness reproduces the papers' numbers (Section 4.5); no design or tuning decision used those runs. Every later change to the protocol is dated and justified in a change log reproduced in Appendix A, which also lists the registration commits with their GitHub push times. Analyses added after the verdicts follow plans committed before they were computed and change no verdict (Appendix H); the variant they point to was then tested in a new pre-registered run (Appendix I). Comparisons use paired tests over tasks: a two-sided Wilcoxon signed-rank test with Holm correction over the declared family (Holm, 1979), and 95% paired-bootstrap intervals with 10,000 resamples. We call a difference significant only if the Holm-adjusted $p < 0.05$ and the interval excludes zero. Every reported number comes from a logged run with its code version, configuration and seed.

4. Diagnosis: How Causal PFNs Fail

Each hypothesis below was registered, with a refutation criterion, before the corresponding runs (Appendix C). All numbers are from the development platform unless a benchmark is named; ratios are $\sqrt{\text{PEHE}}$ relative to the best classical method in the same cell, with paired-bootstrap 95% intervals.

4.1. Accuracy depends on how well the data fit the prior

Against BCF, which is the best classical method in 46 of the 50 cells, CausalPFN is on par in the base design (ratio 1.08

[1.00, 1.16]). It wins clearly only with strong heterogeneity or simple outcome surfaces (effect s.d. 2: 0.67; linear outcome: 0.58) and loses at $d \geq 100$ (1.66 / 1.96), with little heterogeneity (effect s.d. 0: 2.73), under weak overlap ($\kappa = 3$: 1.35) and at $n = 8000$ (1.26 [1.15, 1.38]). Do-PFN and CausalFM are 1.6–5× worse than BCF almost everywhere. They shrink the CATE toward a constant (OLS slope of $\hat{\tau}$ on τ: 0.18 and 0.34 at base, at most 0.15 for $d \geq 25$), their error does not fall with n (0.474 → 0.432 and 0.381 → 0.353 from $n = 250$ to 8000, while CausalPFN goes 0.353 → 0.143), and it grows with the size of the effect (Do-PFN $\sqrt{\text{PEHE}}$ 0.444 at ATE 0, 0.997 at ATE 4; CausalPFN 0.214), as Do-PFN's authors report for their prior; Do-PFN v1 was pretrained on at most 6 features, which its package does not enforce (Table 28). The most accurate estimator on the platform is not a causal PFN. TabPFN-DR, a generic TabPFN inside a cross-fitted DR-learner, is significantly better than the best classical method in 37 of 50 cells and worse in 2; it has the best mean rank (1.44; BCF 2.60, CausalPFN 3.10) and no dimension cliff ($d = 100$: 0.166 vs. BCF 0.271 and CausalPFN 0.451).

4.2. CausalPFN's native uncertainty is aleatoric

CausalPFN's 90% CATE intervals cover 1.00 in all 50 cells; at base their width is 3.64 against a $\sqrt{\text{PEHE}}$ of 0.214. BCF covers 0.87 with intervals 6× narrower; CausalFM is near nominal in prior (0.93) but drops to 0.52–0.85 out of prior. Figure 1a shows why for CausalPFN. The package standardizes Y per arm and was trained with a Gaussian-smoothed histogram loss of width 0.5 in standardized units, so its per-arm variance is

$$\text{Var}\left(\hat{\mu}_t(x)\right) \approx 0.25 \cdot \text{Var}_{\text{ctx}}(Y \mid T = t) + \sigma^2 + r_t, \quad (1)$$

a smoothing floor plus the outcome-noise variance σ^2, with a small remainder r_t that becomes visible only in the hardest cells ($d \geq 100$, random-feature outcomes). The interval is, to first order, a smoothed predictive interval for Y_t, not a posterior for $\mu_t(x)$: at base the per-arm variance is 0.56 while the realized CATE mean squared error is 0.047.

The ATE interval fails in the opposite direction. The ATE posterior is the mean over units of *independent* per-unit draws, so its standard deviation equals $(\sum_i \text{sd}_i^2)^{1/2}/n$ exactly (ratio 1.00 in every cell) and error shared across units is averaged away. Nominal 95% ATE intervals cover 0.55 at base, 0.10 at $d \geq 100$ and 0.00 under hidden confounding, and coverage falls as observed confounding grows (0.95, 0.75, 0.35 at $\rho_c = 0, 0.3, 0.95$; Figure 1b). The package's temperature calibration narrows them further (coverage 0.35): it calibrates factual predictions, already near nominal. Under hidden confounding the native variance even *falls* (1.72 → 1.33) while the CATE error grows 4.4×: the confounder becomes predictable from T, so $\text{Var}(Y \mid T)$ shrinks.

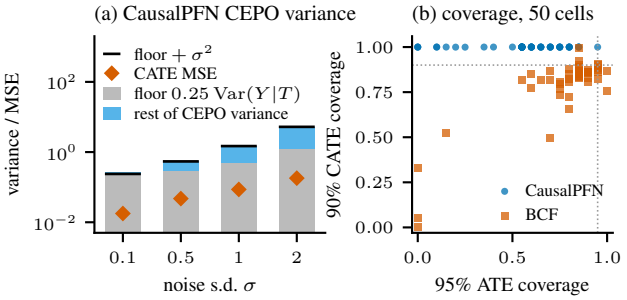


Figure 1. **CausalPFN's uncertainty is aleatoric.** (a) Per-arm CEPO variance (control arm, mean over 20 seeds) split into the smoothing floor $0.25\,\text{Var}(Y \mid T)$ and the rest; the black tick marks floor + σ^2 (Equation (1)) and the diamonds the realized CATE mean squared error, for four noise levels (log scale). (b) Coverage of nominal 90% CATE and 95% ATE intervals across the 50 platform cells for CausalPFN and BCF.

4.3. Native signals do not flag violated prerequisites; the data partly can

None of the native signals detects a hard overlap violation, and under hidden confounding the raw signals move the wrong way, as Equation (1) predicts (Table 4 in Appendix C). Adjusting for the observed within-arm outcome s.d. leaves weak signals; only out-of-prior structure is flagged with some reliability, partly through d. CausalPFN's own calibration statistic does no better (AUROC ≤ 0.50 for hidden confounding and hard overlap).

The data reveal more. The share of units with $\hat{e}$ outside $[0.05, 0.95]$ separates hard and weak overlap violations (AUROC 0.95 and 0.99). A cross-fitted doubly robust estimate of the PFN's excess risk over a reference is unbiased per cell ($d = 150$: +0.215 estimated vs. +0.191 true excess in mean squared CATE error) but noisy per dataset (Spearman 0.70 / 0.67 with the true excess for Do-PFN / CausalFM, and 0.28 for CausalPFN, which is usually close to BCF). Turning the estimate into a decision needs error control. Picking the candidate with the smallest estimated risk is worse than always using CausalPFN (0.705 vs. 0.697 mean $\sqrt{\text{PEHE}}$), whereas switching only when a classical method is better at one-sided $z > 1.645$ costs nothing in prior and recovers 64% of the oracle gain out of prior (0.388 → 0.334; oracle 0.304), with 6% of switches harmful. It misses the moderate gains under weak overlap, where the pseudo-outcomes are heavy-tailed (3% of tasks switch although BCF is 22% better). Hidden confounding stays undetectable by any data statistic (0.27–0.60), as it must. Section 5 builds on these facts.

4.4. Silent limits and a model-level bias

Two limits of the released CausalPFN wrapper act without warning (Figure 3 in Appendix C). For $d > 99$ it projects the covariates onto 99 components with an unsupervised

truncated SVD, which spreads the few active covariates across all components: one extra irrelevant covariate (from $d = 99$ to $d = 100$) raises $\sqrt{\text{PEHE}}$ by $1.41\times$ ($0.320 \to 0.451$; Welch $p = 2 \times 10^{-4}$) while the DR-learner stays flat (0.35). The context is capped at 2048 units per arm per query batch, which binds for $n \gtrsim 3500$; with every unit in context, $\sqrt{\text{PEHE}}$ at $n = 8000$ falls by 14% (ratio 0.86 [0.81, 0.91]) and the loss to BCF disappears ($1.26 \to 1.08$, interval covering 1). Both limits could be fixed in software, but the ranking of Section 4.1 does not rest on them: on the 46 cells they leave untouched, TabPFN-DR is still better than the best classical method in 33 and than CausalPFN in 37 (Appendix H).

The confounding bias, by contrast, is model-level. Under strong observed confounding ($\rho_c = 0.95$, $\kappa = 1.5$), CausalPFN's ATE bias falls like the DR-learner's up to $n = 1000$ and then stays at about 9% of the naive bias ($+0.084 \pm 0.006$ at $n = 8000$, coverage 0.00); CausalFM keeps about 24% at every n, while the AIPW and BCF biases keep shrinking ($+0.030$, $+0.014$). Full context leaves the bias unchanged ($+0.084$), so the cap is not the cause. A plug-in TabPFN T-learner halves its bias with every doubling of n, and TabPFN-DR removes it ($+0.000 \pm 0.008$ at $n = 2000$).

4.5. Evaluation choices decide the winner

Our harness reproduces the published CausalPFN PEHE on IHDP, ACIC-2016 (10 datasets) and both LaLonde samples within one standard error, and CausalFM / Do-PFN on Jobs within 4% (Appendix J). On the same 10 ACIC-2016 datasets, in the normalized units of CausalFM's evaluation code, our CausalFM and Do-PFN errors (0.619, 0.789) match those published by CausalFM's authors (0.638, 0.857; Table 27), so their poor results are not an artefact of how we call them. The same table reports a DR-learner at 1.485 in these units, against 0.156 for ours, so the comparison there favours the PFNs. The audit found scoring problems. CausalPFN's IHDP loader scores the ATE of all 100 realizations against a constant 4, although the true values range from -0.60 to 12.65. This reproduces the published relative ATE error (0.203 ± 0.038); the correct truth gives 0.0233 ± 0.0027, $8.7\times$ smaller, and moves the causal forest from the best to the worst classical method. The LaLonde files contain only noisy potential outcomes, and the common ACIC-2016 copy has 10 of the 77 settings. The benchmark then decides the winner: the CausalPFN/BCF ratio is 0.22 on IHDP and 1.30 on the ACIC-10 copy, where TabPFN-T is best.

5. Certify-or-Defer

Section 4 leaves a user with a frozen PFN that may be excellent or badly wrong on their data, and with native signals that cannot tell the two apart. Certify-or-defer answers the question a user can act on: *is the PFN's CATE worse than a classical reference on this dataset, and if so, where?* It wraps any causal PFN without retraining and uses only (X, T, Y).

5.1. In-context cross-fitting and a paired excess-risk statistic

Split $\mathcal{D}$ into K folds $I_1, \dots, I_K$, stratified on T. A PFN needs no refitting: for fold k we pass the context $\mathcal{D}_{-k}$ and query X_i, $i \in I_k$, which gives out-of-fold estimates $g(X_i) := \hat{\tau}_f[\mathcal{D}_{-k}](X_i)$ from K forward passes. Each reference learner $r \in \mathcal{R}$ is fit on $\mathcal{D}_{-k}$ and predicts fold k, and so are the scoring nuisances $\eta = (\hat{e}, \hat{m}_0, \hat{m}_1)$. The doubly robust pseudo-outcome

$$\varphi_i = \hat{m}_1 - \hat{m}_0 + \frac{T_i(Y_i - \hat{m}_1)}{\hat{e}} - \frac{(1 - T_i)(Y_i - \hat{m}_0)}{1 - \hat{e}} \quad (2)$$

(all nuisances evaluated at X_i, $\hat{e}$ clipped to $[c, 1 - c]$) has $\mathbb{E}[\varphi \mid X] = \tau(X)$ when the nuisances are correct (Robins et al., 1994; Kennedy, 2023). We compare f with r through the paired statistic

$$\delta_i = \big(g - r\big)(X_i)\,\big[(g + r)(X_i) - 2\varphi_i\big], \quad (3)$$

which equals $(g(X_i) - \varphi_i)^2 - (r(X_i) - \varphi_i)^2$, with $\hat{\Delta} = \frac{1}{n}\sum_i \delta_i$, $\hat{\sigma}^2$ its sample variance and $Z = \sqrt{n}\,\hat{\Delta}/\hat{\sigma}$. A positive $\hat{\Delta}$ says that f has larger CATE risk than r.

Two properties make Equation (3) usable on a single dataset. First, the unknown τ^2 cancels, and the outcome noise enters only through the factor $(g - r)$, so $\mathrm{Var}(\delta)$ scales with $\|g - r\|_\infty^2 \,\mathrm{Var}(\varphi)$ instead of the $\mathrm{Var}(\varphi^2)$ of two separately estimated risks. Second, with $b_k(x) = \mathbb{E}[\varphi(O; \hat{\eta}_k) - \tau(X) \mid X = x, \mathcal{D}_{-k}]$,

$$\mathbb{E}[\delta_i \mid \mathcal{D}_{-k}] = \underbrace{R_k(g) - R_k(r)}_{\Delta_k} - 2\,\mathbb{E}_X\big[(g - r)(X)\, b_k(X)\big], \quad (4)$$

where $R_k(h) = \mathbb{E}_X[(h(X) - \tau(X))^2 \mid \mathcal{D}_{-k}]$. For the AIPW score, $b_k = (e - \hat{e}_k)\big[(\mu_1 - \hat{m}_{1k})/\hat{e}_k + (\mu_0 - \hat{m}_{0k})/(1 - \hat{e}_k)\big]$, a product of nuisance errors, so the bias is second order, provided that $\hat{e}_k$ can approach e; where $e < c$, clipping prevents this and the bias becomes first order (Table 2). Nothing is assumed about the PFN or its prior: a misspecified PFN is detected, not trusted. The target is $\bar{\Delta} = \frac{1}{K}\sum_k \Delta_k$, the excess risk at context size $(1 - 1/K)n$.

Assumption A1 (Overlap). $e(x) \in [c, 1 - c]$ for the clipping constant c.

Assumption A2 (Boundedness). $|Y|, |g|, |r| \le M$.

Assumption A3 (Product rate). $\|e - \hat{e}_k\|_2 \max_t \|\mu_t - \hat{m}_{tk}\|_2 = o_p(n^{-1/2})$.

Assumption A4 (Non-degeneracy). $\mathrm{Var}(\delta_i \mid \mathcal{D}_{-k}) \ge \sigma_0^2 > 0$.

Assumption A5 (Rank stability). $\operatorname{sign}(R_n(f) - R_n(r)) = \operatorname{sign}(\bar{\Delta})$ whenever $|\bar{\Delta}| \geq \epsilon_n \to 0$, overall and within each region used below.

Assumption A5 is the usual premise of cross-validation: comparing f and r at $(1-1/K)n$ units ranks them correctly at n units. Assumption A4 excludes $g \approx r$, where deferring would change nothing.

Proposition 5.1 (Asymptotic normality). *Under Assumptions A1 to A4 and a stability condition on the fold-wise fits (Assumption A6, Appendix D),* $\sqrt{n}(\hat{\Delta} - \bar{\Delta})/\hat{\sigma} \Rightarrow \mathcal{N}(0,1)$.

5.2. Region-wise do-no-harm deferral

Testing on the whole sample loses power exactly where Section 4.3 found it lacking: under weak overlap, φ is heavy-tailed. We therefore split the units by their out-of-fold propensity into $G_{\text{in}} = \{\hat{e} \in [0.1, 0.9]\}$ and its complement G_{out}, and test $H_0^{g,r} : \bar{\Delta}_{g,r} \leq 0$ for every region g and reference r, with one-sided normal p-values from $Z_{g,r}$ and Holm's procedure over all $|\{\text{in}, \text{out}\}| \times |\mathcal{R}|$ tests at level α. In each region the output is f unless one of that region's hypotheses is rejected, in which case it is the rejected reference with the largest $\hat{\Delta}_{g,r}$ (Algorithm 1, Appendix E). Test units are assigned to regions by their propensity estimate. This is the configuration we evaluate, A-V5 (variant V5 in Appendix E); A-V5-R, which we recommend after the verdicts, uses the R-loss score instead of Equation (3) (Appendix I).

Proposition 5.2 (Do-no-harm deferral). *Under the conditions of Proposition 5.1 for every region and reference: (i)* size*: if $\bar{\Delta}_{g,r} \leq 0$ for all (g, r), the probability that certify-or-defer defers anywhere is at most $\alpha + o(1)$; (ii)* power*: if $\bar{\Delta}_{g,r} \geq \delta_0 > 0$ for some (g, r), it defers in region g with probability $\to 1$; (iii)* risk*: with Assumption A5 and regions estimated consistently, the output $\hat{h}$ satisfies $\mathbb{E}[R_n(\hat{h})] \leq R_n(f) + \alpha \sum_g w_g \max_r \left(R_{n,g}(r) - R_{n,g}(f)\right)_+ + o(1)$, where w_g is the mass of region g and $R_{n,g}$ the risk on it; if in some region a unique reference is strictly best and better than f, $\hat{h}$ uses it there with probability $\to 1$.*

Holm controls the family-wise error of the asymptotically valid tests of Proposition 5.1; proofs are in Appendix D. The guarantee is a do-no-harm guarantee relative to f: certify-or-defer never needs the PFN to be calibrated, and it does not promise to find the best estimator. Assumptions A1 and A2 exclude hard overlap violations, which include about a tenth of the units of ACIC-2016, and heavy-tailed outcomes (H3); Appendix H measures the size there.

5.3. What the certificate reports, and what it cannot

Alongside the output, certify-or-defer reports the full-sample Z_{DR} against the DR-learner, the share of units with $\hat{e} \notin [0.05, 0.95]$, and the best-linear-projection slope of φ on the centred PFN estimate (Chernozhukov et al., 2017), which is far above 1 when f shrinks heterogeneity. It also returns an AIPW estimate of the ATE from f's out-of-fold estimates (Appendix E), reported descriptively. A certificate, i.e. no rejection, says only that the data show no excess risk of f over the references; it is not evidence that f is accurate. If f and every reference are equally poor, f is kept. Hidden confounding violates identification rather than Assumptions A1 to A5: φ is then centred on a confounded contrast and no statistic of $P(X, T, Y)$ can reveal it. The certificate says so and points to sensitivity analysis (Javurek et al., 2026).

Configuration and cost. We use $\mathcal{R} = \{\text{DR-learner}, \text{BCF}\}$ with the baseline configurations, $K = 5$, $c = 0.01$, $\alpha = 0.05$, and scoring nuisances that average LightGBM and a random forest (Breiman, 2001), so they differ from the DR-learner's own; all fixed on development data (Section 6.1). The cost is $K + 1$ PFN passes plus a reference bundle shared across PFNs, which dominates.

6. Pre-Registered Development and Confirmatory Evaluation

6.1. Pilots on development data

The method was fixed in two pre-registered pilots on twelve platform cells that span in-prior and failure regimes (Appendix F), with 20 seeds each. A *failure event* is a task on which the PFN's $\sqrt{\text{PEHE}}$ exceeds the DR-learner's by at least 10%. The criteria, set before the first run, were: **P-1** the AUROC of Z_{DR} for failure events is at least 0.80 and exceeds the best native signal (raw or variance-adjusted CATE s.d.) with a paired-bootstrap interval excluding 0; **P-2** the output is within 2% of always-PFN in every in-prior cell, recovers at least half of the oracle gain over the failure cells, and at most 10% of deferrals are harmful; **P-3** the AIPW ATE interval covers at least 0.90 in every cell except hidden confounding. Passing required all criteria for both PFNs.

In P01 (seeds 0–19), detection passed for both PFNs, but no single variant passed P-2 for both (the single-region test failed CausalFM's harm bound at 0.12; the region-wise test missed CausalPFN's gain threshold by 0.0002), and the ATE interval undercovered for both, as did classical AIPW. We dropped the ATE coverage claim, narrowed the method to the region-wise variant and replicated it on fresh seeds 20–39 (P02) with unchanged thresholds. In P02 the deferral claim replicated for both PFNs, and CausalPFN's output is never more than 1.8% worse in prior. Detection failed for CausalPFN on superiority, not accuracy: Z_{DR} reaches AUROC 0.978, but 72 of its 76 failure events are at $d \geq 100$, where its adjusted native s.d. also flags them (0.958). The verdict was a fail (Table 8 in Appendix F). As planned, A-V5 went unchanged to the confirmatory test, whose criterion

predates the pilots.

6.2. Final evaluation

Criterion. The success criterion, frozen with the protocol, has four parts. **A-1 (detection):** on H and on B2, the AUROC of Z_{DR} for failure events is at least 0.80 and exceeds the best native signal of the same PFN (paired bootstrap, Holm over $\{\mathrm{H}, \mathrm{B2}\} \times \{\mathrm{CausalPFN}, \mathrm{CausalFM}\}$). **A-2 (do-no-harm):** on H and B2, mean $\sqrt{\mathrm{PEHE}}$ of the output is no larger than always-PFN, significantly lower on at least one of them, and at most 10% of the deferring tasks, pooled over H and B2, are harmed. **Non-inferiority:** on each of B1–B4, the one-sided 95% upper bound of the $\sqrt{\mathrm{PEHE}}$ ratio to always-PFN is at most 1.02. **Generality:** A-1 and A-2 hold for both PFNs. The ATE coverage of the AIPW output is reported descriptively. Baselines are always-PFN, always-reference, the naive argmin of the estimated risks, causal isotonic calibration of the PFN (van der Laan et al., 2023), TabPFN-DR, BCF, causal forest and, for the ATE, MP-OSPC.

Scope. Every method runs on every task, with two runtime exceptions fixed in advance: MP-OSPC runs on a predeclared 77-task subset of B2 (one replication per setting), and no TabPFN-based method runs on B3, whose 14.5k training units exceed TabPFN v2's size limit (Appendix A).

Results. The final run has 13,517 job records, 0 of them not completed (consistency checks in Appendix G). Table 1 gives the verdict on every pre-registered claim, and Figure 2 the accuracy of A-V5 and the main baselines relative to always-PFN.

Detection (A-1) fails for CausalPFN and fails for CausalFM. On H, Z_{DR} separates failure events well for both PFNs (AUROC 0.944 and 0.859). On B2 it falls short of 0.80 for both (0.783 and 0.788); for CausalFM, with failure events on 746/770 tasks, its margin over the native signal is not significant either. As on the platform (Section 4.3), the excess-risk estimate is noisy per dataset, and many B2 tasks lie near the 10% threshold (Table 22).

Do-no-harm (A-2) fails for CausalPFN and passes for CausalFM. Pooled over H and B2, 18.9% and 3.3% of the deferring tasks are made worse (bound 10%). Mean $\sqrt{\mathrm{PEHE}}$ is significantly lower than always-PFN on both benchmarks for both PFNs, so CausalPFN's verdict is decided by the harm bound alone. The harms are concentrated (Table 22): for CausalPFN on B2, 65 of 97 come from tasks deferred to the DR-learner in G_{out} only (Table 2); for CausalFM on H, 17 of 19 occur under hidden confounding (H6), which no statistic of (X, T, Y) can reveal.

Non-inferiority on B1–B4 passes for CausalPFN and passes for CausalFM, and generality fails.

Against the strongest baselines. TabPFN-DR is more accurate than A-V5 for both PFNs on every benchmark it runs on, except B1 with CausalPFN (paired differences in Appendix G). Against BCF, A-V5 with CausalPFN is never worse and ties on B2 and H; with CausalFM it is worse on H. The naive argmin is more accurate than A-V5 on H and B2 for CausalPFN ($\sqrt{\mathrm{PEHE}}$ 1.174 vs. 1.220 on B2): on H the test lowers the share of harmful switches from 19% to 6%, on B2 it does not (22% vs. 21%).

ATE and seeds (descriptive). The 95% AIPW ATE interval of A-V5 covers 0.99 on B1 and 0.85 on B2 for CausalPFN, against 0.52 for CausalPFN's own interval on B2 (CausalFM: 0.68 and 0.64); under H5 and H6, where ignorability given X fails, it covers 0 (Table 19). Across method seeds, its decisions agree on $\geq$ 99% of B1 and 83–84% of H tasks at $n = 500$ (Table 21).

Post-verdict analyses and a confirmatory run. Planned before they were computed and unable to change any verdict, the analyses of Appendix H use the true CATE to locate the test's errors (Table 2): CausalPFN's harm comes from a bias of the DR score where overlap fails. We recommend the R-loss score (A-V5-R), which the development data alone would also favour (harm 4.3% against 8.6% for CausalPFN), and tested it in a run pre-registered on new data, ACIC-2017 (Appendix I). It passed both conditions for both PFNs (error-ratio bound 0.768 and 0.401; harm 1/129 and 0/122). The registered A-V5 did no harm there either and was more accurate (0.427 vs. 0.489 for CausalPFN), and always-BCF more so (0.341): A-V5-R does no harm on new data, but the run cannot separate it from A-V5.

7. Discussion and Limitations

What the confirmatory test showed. Certify-or-defer worked where the diagnosis says a data-driven check is strong and failed where it says such checks are weak. It never lost accuracy, and it repaired CausalFM's large errors, where the paired test has power, within the harm bound. It failed where differences are small, which caps detection on ACIC-2016, and where overlap fails: there the clipped propensity makes the bias of the doubly robust score first-order, and the bias favours the DR-learner it judges (Curth & van der Schaar, 2023). On ACIC-2017 overlap fails on half of the tasks, yet the DR score showed no such bias (median -0.15 SE); in R2-A the clipping bias also scales with the outcome-model error.

What we recommend. On tabular CATE tasks, a generic in-context regressor inside an orthogonal estimator (TabPFN-DR) is a better default than a causal PFN, wrapped or not, and causal pretraining should be judged against it; our references were fixed before the development

Table 1. Pre-registered claims on the final benchmarks. A-1: AUROC of Z_{DR} / best native signal for failure events, paired-bootstrap 95% interval of the difference, Holm-adjusted p; pass: AUROC ≥ 0.80, interval > 0, $p < 0.05$. A-2: mean $\sqrt{\mathrm{PEHE}}$ always-PFN $\rightarrow$ A-V5, interval of the difference, Holm-adjusted Wilcoxon p, share of tasks with a deferral / share of those made worse (the criterion pools H and B2). NI: ratio of mean $\sqrt{\mathrm{PEHE}}$ to always-PFN (one-sided 95% upper bound; margin 1.02).

		CausalPFN				CausalFM			
A-1	set	AUROC Z / native	95% CI	p		AUROC Z / native	95% CI	p	
	H	0.944 / 0.555	[+0.307, +0.471]	0.0004	✓	0.859 / 0.650	[+0.126, +0.290]	0.0004	✓
	B2	0.783 / 0.591	[+0.143, +0.240]	0.0004	×	0.788 / 0.646	[−0.006, +0.287]	0.0624	×
A-2	set	$\sqrt{\mathrm{PEHE}}$	95% CI	p	def./harm	$\sqrt{\mathrm{PEHE}}$	95% CI	p	def./harm
	H	2.130 → 1.901	[−0.361, −0.140]	<0.0001	30% / 6%	3.839 → 2.562	[−1.607, −0.964]	<0.0001	61% / 13%
	B2	1.424 → 1.220	[−0.235, −0.174]	<0.0001	60% / 21%	4.227 → 1.196	[−3.148, −2.915]	<0.0001	100% / 1%
NI		B1	B2	B3	B4	B1	B2	B3	B4
	ratio (upper)	0.999 (1.000) ✓	0.857 (0.875) ✓	1.002 (1.004) ✓	0.996 (0.999) ✓	0.359 (0.381) ✓	0.283 (0.297) ✓	0.739 (0.743) ✓	0.660 (0.676) ✓
overall generality		A-1 ×, A-2 ×, NI ✓				A-1 ×, A-2 ✓, NI ✓			
		× (A-1 and A-2 for both PFNs)							

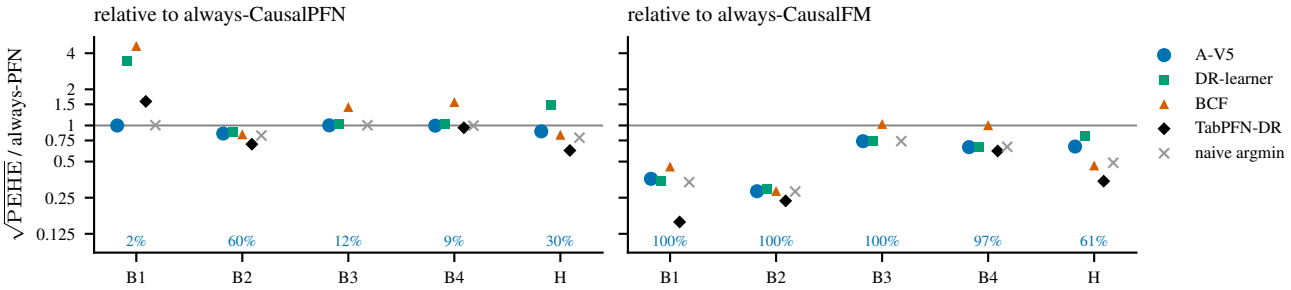


Figure 2. **Accuracy on the final benchmarks.** Mean $\sqrt{\mathrm{PEHE}}$ of A-V5, the DR-learner, BCF, TabPFN-DR and the naive argmin, divided by that of the PFN it wraps (left CausalPFN, right CausalFM; log scale, below 1 is better). Percentages: share of tasks on which A-V5 deferred in some region. On B3 / B4 $\sqrt{\mathrm{PEHE}}$ is an ITE-RMSE against noisy ITEs; TabPFN-DR is not run on B3.

Table 2. Post-verdict analyses (Appendix H) and F02 (Appendix I). B2: CausalPFN, G_{out} vs. DR-learner.

analysis	finding
Size (R-B)	B2: 109/249 true nulls rejected (82/148 at a 1-SE margin); 80 of 97 harmful deferrals follow one; harm 65/79 at oracle excess ≤ 0, 5/152 above 3 SE.
Bias (R2-A)	B2: 2 rejections without the bias; median bias +2.95 SE, mostly from clipping $\hat{e}$ where $e < 0.01$; it favours the DR-learner.
Scores (R-A)	Pooled harm: R-loss 9.8%, G_{in} only 7.6%, OW-DR 18.3% (registered 18.9%); naive argmin 1.226× the PFN's error on B1.
Splits (R-C)	10 splits: agreement 83% → 92%; H harm 5.6% → 1.8%; B2 unchanged.
F02	ACIC-2017: A-V5-R confirmed (ratio bound 0.768 / 0.401, harm 1/129 / 0/122); A-V5 also harmless and more accurate, always-BCF more so.

data showed this (Section 4.1). Certify-or-defer serves a user who keeps a PFN: it bounds the loss where the PFN is good, which the naive argmin does not, at 8.8–27× the PFN's cost. TabPFN-DR costs more at scale (median per B2 task 4,463 s against 1,246 s; on H 187 against 357 s).

What the guarantees cover. Proposition 5.2 is asymptotic, assumes bounded overlap and outcomes, reaches the full-data output only through rank stability (Assumption A5), and bounds the expected excess risk, not the harm share that A-2 counts. Its size held for CausalPFN on H and the platform but not on ACIC-2016, whose low-overlap region violates Assumption A1; for CausalFM it failed mainly under hidden confounding and hard overlap violations, which the assumptions also exclude.

Pre-registration. Registration is internal; the GitHub push times of its commits precede the runs (Appendix A) and become checkable once the repository is released. Two diagnostic refutation criteria (D1, D6) were weak, the variant was chosen after the first pilot and A-V5-R after the verdicts, so only the confirmatory claims were fixed before any data informed them.

Implications for causal PFNs. Report epistemic uncertainty apart from the spread of Y_t, form the ATE from joint draws, and warn when preprocessing changes the input, the dimension exceeds the pretraining range or overlap is weak.

Impact Statement

This paper studies when amortized causal-effect estimators can be relied on and proposes a check that falls back to classical estimators when they cannot. Effect estimates inform decisions in medicine, policy and business, so silent failures of such estimators can cause harm; making those failures visible is the intended benefit. The check cannot detect hidden confounding, and a passed certificate must not be read as evidence that the PFN is accurate or that the identifying assumptions hold. All data are public benchmark or synthetic data; no new human-subject data were collected.

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A. Protocol and Change Log

The protocol (v1) was drafted on 2026-10-08 before any method was designed and frozen on the same day, after sign-off of the method proposal and before any of its code was run. The full text, with the change log, is part of the supplementary material; we summarize it here.

Benchmarks. B1: IHDP-100 (`ihdp_npci_1-100`, 672 training / 75 test units); the ATE target is the mean CATE over the training units, not the constant stored with the data. B2: ACIC-2016, all 77 settings $\times$ 10 replications generated with the official `aciccomp2016` package (4802 units, random 90/10 split seeded by setting and replication; categorical covariates ordinal-coded). B3 / B4: the 100 RealCause LaLonde CPS / PSID samples shipped with the CausalPFN repository (90/10 split); the truth is a noisy ITE, so ATE metrics are primary and no CATE coverage is reported. H: Appendix B. Also reported, outside the success criterion: the CausalPFN paper's own protocol on B1–B4, to connect with published numbers (Appendix J).

Metrics and statistics. $\sqrt{\text{PEHE}}$ on test units (primary CATE metric), |ATE error| (primary ATE metric), coverage and width of 90% CATE and 95% ATE intervals where the truth is noiseless, detection AUROC, wall time and peak memory. Paired comparisons over tasks: two-sided Wilcoxon signed-rank tests with Holm correction within the declared family, and 95% paired-bootstrap intervals (10,000 resamples, tasks resampled jointly). An improvement is claimed only if the Holm-adjusted $p < 0.05$ and the interval excludes 0.

Methods and tuning. CausalPFN 0.1.5 with the released weights (package defaults; temperature calibration with the package's own procedure), Do-PFN v1 (released artifacts), CausalFM (released checkpoint, inputs standardized on observed data), TabPFN v2.0.9 as T-learner and inside the DR-learner (5-fold cross-fitting, propensity clipped at 0.01), MP-OSPC (authors' code and defaults), DR-learner (EconML 0.17, LightGBM 4.7 nuisances tuned by 3-fold grid search over 16 configurations, 5-fold cross-fitting, clip 0.01), causal forest (EconML `CausalForestDML`, 1000 trees, minimum leaf size 5, the same nuisances), BCF (stochtree, 10 grow-from-root and 500 MCMC iterations, cross-fitted tuned LightGBM propensity). The PFNs are used as released; for CausalPFN this includes the SVD for $d > 99$ and the context cap of Section 4.4.

Change log (dated entries recorded before the runs they concern).

- 2026-10-08: TabPFN-based methods omitted on B3 (about 16k rows, above TabPFN v2's 10k pretraining limit; CPU cost about 1 h per task); decided from runtime before any B3 TabPFN result existed.
- 2026-10-08 (sign-off): the LaLonde baselines keep the frozen configurations even though the causal forest and BCF were seen to fail there during reproduction; failures are reported as they are, with a per-benchmark sanity column (each method's |ATE err| next to the naive difference in means). No GPU is used.
- 2026-10-09, before any final-evaluation run: the method is fixed to A-V5 after the pilots (Section 6.1); the ATE coverage claim becomes descriptive; the method set and the runtime-based scope (MP-OSPC omitted on B3; TabPFN-DR and MP-OSPC on a pre-declared 77-task subset of B2); TabPFN-DR and BCF as the two baselines of the seed-robustness re-run, chosen by their mean rank on the development platform; each job runs in a fresh process, so wall time includes model loading for every method alike, and no separate warm-up run is made.
- 2026-10-09, during the final run, before any final result was inspected: TabPFN-DR runs on all 770 B2 tasks rather than the subset (runtime-only restriction lifted after the development platform showed TabPFN-DR to be the strongest baseline). The extra tasks run from a code version that differs from the launch version only by this entry and a driver script.
- 2026-10-09, during the final run, before any final result was inspected: a written analysis plan fixes the choices the protocol leaves open. The harmful-deferral rate is pooled over H and B2 (per-benchmark rates are also reported); the bootstrap p-value of an AUROC difference is two-sided, $2\min\{\mathbb{P}(\text{diff} \le 0), \mathbb{P}(\text{diff} \ge 0)\}$; the native signal is the PFN's mean CATE s.d., raw or adjusted for the within-arm outcome s.d. with the adjustment fitted on development cells, in its most favourable orientation, as in the pilots; and the non-inferiority bound is a one-sided 95% paired-bootstrap bound on the ratio of mean $\sqrt{\text{PEHE}}$.

- 2026-10-10, during the final run, before any final result was inspected: the run was paused to free the shared server and later resumed from the same frozen code versions. Jobs in progress at the pause wrote no record and were run again from the start; no completed job was rerun. Wall times of the B2 TabPFN-DR and MP-OSPC jobs therefore mix load levels.
- 2026-10-13, after the verdicts: analyses requested in review, planned before they were computed (Appendix H); exploratory, no verdict changes.
- 2026-10-14, after a second review: analyses R2-A, R2-B and R2-E, planned before they were computed (Appendix H); exploratory, no verdict changes.
- 2026-10-14, before any method ran on ACIC-2017: the protocol of F02, the confirmatory run of A-V5-R, with its data, criterion, driver and analysis script (Appendix I).
- 2026-10-15, after F02 finished and before its analysis: the run's summary file, which the driver's last call had overwritten, was rebuilt from the saved job records with the harness's collector; no frozen file changed.

Evidence for the registration dates. The registration is internal to our repository. Its commits also carry GitHub's server-side push times, which the commit author cannot set (retrieved from the repository's event log on 2026-10-13, archived with the code): the held-out suite H (ad186f3) at 2026-10-08 04:42 UTC, the frozen protocol with the method specification and the P01 plan (83f809d) at 13:54, the P02 launch version (fee2124) at 21:25, the final-evaluation launch version (5b04464) at 23:15 and its analysis plan (257e2d2) at 23:17, before the first final-evaluation record was written. The later registrations, retrieved on 2026-10-15, are the first review's analysis plan (ee56bd9) at 2026-10-13 05:19 UTC, the second review's plan (1470599) at 2026-10-14 20:36, and the F02 protocol with its driver and analysis script (786cc4f) at 21:16, 24 seconds before F02 was launched; its first job record was written at 21:31. They become checkable by others once the repository is released with its history.

Deviations. The diagnosis plan specified 30 seeds per cell unless stated; the platform sweeps were registered with 20 seeds in the same commit, and the bias-versus-n study (E03) used 30.

B. Development Platform and Held-Out Suite

Platform. Covariates are Gaussian by default (correlated, mixed or binary in the covariate sweep). The prognostic function f, the effect function g and an auxiliary function q are random functions of the first 5 covariates, each normalized to mean 0 and variance 1 on a reference sample. Treatment follows $\operatorname{logit} \mathbb{P}(T=1 \mid X,U) = \kappa\,(\rho_c f(X) + \sqrt{1-\rho_c^2}\,q(X)) + b + \gamma_u U$, with b set so that the treated share is 0.4; under a hard overlap violation, units whose first covariate exceeds a quantile are always treated. Outcomes are $Y_0 = s\,(f(X) + \delta_u U + \varepsilon) + c_0$ and $Y_1 = Y_0 + s\,\tau(X)$ with $\tau = \mathrm{ATE} + h\,g(X)$, scale s and shift c_0. The base design has $n = 1000$ training and 500 test units, $d = 10$, additive f, linear g, ATE 0.5, $h = 0.5$, $\kappa = 1$, $\rho_c = 0.6$, $\gamma_u = \delta_u = 0$ and Gaussian ε with s.d. 0.5. The one-factor-at-a-time sweeps are: outcome function (linear, interaction, random features), effect function (additive, interaction, random features), ATE $\in \{0, 0.25, 1, 2, 4\}$, $h \in \{0, 0.25, 1, 2\}$, noise family (Laplace, heteroskedastic), noise s.d. $\in \{0.1, 1, 2\}$, covariates (correlated, mixed, binary), $s \in \{0.01, 100\}$, $c_0 = 100$, $d \in \{5, 25, 50, 100, 150\}$, $n \in \{250, 500, 2000, 4000, 8000\}$, $\kappa \in \{0.25, 0.5, 2, 3\}$, $\rho_c \in \{0, 0.3, 0.95\}$, share of always-treated units $\in \{0.05, 0.1, 0.2\}$, and hidden confounding $\gamma_u \in \{0.5, 1, 2\}$ with $\delta_u = 1$. The noise-matched control for hidden confounding keeps $\delta_u = 1$ but sets $\gamma_u = 0$, so U adds the same outcome variance without confounding.

Held-out suite H. Written on 2026-10-08 before any method design and frozen; only data-validity checks were run on it before the final evaluation. Each generator lies outside the platform's axes and, in at least one respect, outside the published priors. H1 (Kang–Schafer): observed covariates are nonlinear transforms of latent normals, so both nuisances are misspecified in the observed space, with a heterogeneous effect added (Kang & Schafer, 2007). H2: Friedman-#1 prognosis on $U[0,1]^d$ covariates with a step-function CATE (Friedman, 1991). H3: a log-normal positive outcome with a multiplicative treatment effect and t_3 noise. H4: a 20-level categorical covariate with group random effects in prognosis and effect. H5: a pre-treatment collider M (with $U_1 \to T$, $U_2 \to Y$, $U_1, U_2 \to M$) among the covariates. H6: strong instruments in X plus moderate hidden confounding (bias amplification). The truth $\mu_t(x) = \mathbb{E}[Y_t \mid X = x]$ is computed analytically or by Monte Carlo over the latent variables. Each generator is run at $n \in \{500, 2000\}$ with 20 seeds and 500 test units (240 tasks).

C. Diagnosis Details

Table 3. Pre-registered diagnostic hypotheses (written before the runs that test them) and their verdicts.

	hypothesis	refuted if	verdict
D1	PFN accuracy relative to classical estimators degrades along axes that move the data away from the prior	relative $\sqrt{\text{PEHE}}$ stays within $\pm 10\%$ of its base value on all axes	not refuted (d, effect s.d., ATE level)
D2	under strong observed confounding the PFN ATE bias does not vanish as fast as AIPW's	PFN $\lvert\text{bias}\rvert$ falls at least as fast as AIPW's and is within 2 s.e. of 0 at $n \geq 4000$	not refuted for all three PFNs
D3	CATE intervals undercover out of prior; CausalPFN's ATE intervals undercover even in prior (independent draws)	coverage within nominal bands in every cell, or $\text{ICE}_\mu \leq \text{ICE}_\tau$ in $\geq 95\%$ of tasks incl. stress cells	ATE half confirmed; CATE half refuted for CausalPFN (over-coverage), confirmed for CausalFM
D4	native signals do not discriminate violated from clean datasets	a native signal reaches AUROC ≥ 0.80 and Spearman ≥ 0.5 for a testable violation	not refuted
D5	classical estimators catch up as n and d grow	the best PFN stays significantly better at the largest n and d	not refuted
D6	evaluation choices change conclusions	all reproductions within 2 s.e. and rankings unchanged by metric / truth choices	not refuted (constant IHDP truth; benchmark-dependent winner)
D7	benchmark rankings do not transfer to the platform	Kendall $\tau \geq 0.6$ in most cells	refuted ($\tau \geq 0.6$ in 76% / 82% of cells for IHDP / ACIC)
S1	overlap and out-of-prior failures are observable from (X, T, Y)	no data statistic reaches AUROC ≥ 0.80 (CI lower bound > 0.65) where native signals are < 0.75	overlap: solvable; structure: partly; hidden confounding: not testable (expected)
S2	the bias plateau and the large-n loss are caused by the context cap	full-context bias within 2 s.e. of the default at $n = 4000, 8000$	bias: refuted (model-level); large-n loss: confirmed

Table 4. Detection of violated prerequisites (dataset-level AUROC, violated vs. reference cells; hidden confounding vs. a noise-matched control). "adj.": adjusted for the observed within-arm outcome s.d. Data statistics use only (X, T, Y): overlap share = share of units with $\hat{e} \notin [0.05, 0.95]$; DR excess = cross-fitted DR estimate of the PFN's excess risk over the DR-learner.

signal	hidden conf.	hard overlap	weak overlap	out-of-prior
CausalPFN CATE s.d.	0.10	0.53	0.38	0.85
adj.	0.68	0.48	0.71	0.66
CausalFM CATE s.d., adj.	0.55	0.51	0.64	0.87
Do-PFN width, adj.	0.73	0.47	0.64	0.75
overlap share	0.30	**0.95**	**0.99**	0.31
DR excess, CausalPFN	0.49	0.51	0.43	0.56
DR excess, CausalFM	0.58	0.45	0.50	0.72

Table 5. Development platform, $\sqrt{\text{PEHE}}$ per cell (mean over 20 seeds) and, in parentheses, the ratio to the best classical method of the cell (paired-bootstrap CIs in the source file). CausalPFN-cal has the same point estimates as CausalPFN. src: `runs/e02_platform/analysis/d1_accuracy.md`.

cell	best cl.	CausalPFN	Do-PFN	CausalFM	TabPFN-T	TabPFN-DR	DR	CF	BCF
base	bcf	0.214 (1.08)	0.445 (2.25)	0.368 (1.86)	0.231 (1.17)	0.163 (0.82)	0.285 (1.44)	0.277 (1.40)	0.198
ate=0.0	bcf	0.214 (1.07)	0.444 (2.22)	0.396 (1.98)	0.231 (1.15)	0.163 (0.81)	0.285 (1.43)	0.275 (1.37)	0.200
ate=0.25	bcf	0.214 (1.08)	0.438 (2.21)	0.380 (1.91)	0.231 (1.16)	0.163 (0.82)	0.284 (1.43)	0.275 (1.38)	0.199
ate=1.0	bcf	0.214 (1.09)	0.480 (2.44)	0.365 (1.85)	0.231 (1.17)	0.163 (0.83)	0.285 (1.45)	0.277 (1.41)	0.197
ate=2.0	bcf	0.214 (1.10)	0.601 (3.10)	0.435 (2.25)	0.231 (1.19)	0.163 (0.84)	0.285 (1.47)	0.282 (1.46)	0.194
ate=4.0	bcf	0.214 (1.08)	0.997 (5.02)	0.750 (3.78)	0.231 (1.16)	0.163 (0.82)	0.285 (1.44)	0.295 (1.49)	0.198
covariates=binary	bcf	0.125 (1.15)	0.440 (4.05)	0.368 (3.39)	0.112 (1.03)	0.112 (1.03)	0.174 (1.60)	0.180 (1.66)	0.109
covariates=corr	bcf	0.203 (0.96)	0.411 (1.94)	0.364 (1.72)	0.235 (1.11)	0.162 (0.77)	0.306 (1.45)	0.270 (1.27)	0.212
covariates=mixed	bcf	0.183 (1.05)	0.434 (2.49)	0.397 (2.28)	0.185 (1.06)	0.144 (0.83)	0.254 (1.46)	0.249 (1.43)	0.174
d=100	bcf	0.451 (1.66)	0.689 (2.54)	0.496 (1.83)	0.256 (0.95)	0.166 (0.61)	0.347 (1.28)	0.367 (1.35)	0.271
d=150	bcf	0.571 (1.96)	0.675 (2.32)	0.510 (1.75)	0.252 (0.87)	0.178 (0.61)	0.373 (1.28)	0.392 (1.35)	0.291
d=25	bcf	0.269 (1.14)	0.523 (2.21)	0.453 (1.91)	0.238 (1.01)	0.171 (0.72)	0.318 (1.34)	0.327 (1.38)	0.237
d=5	bcf	0.188 (1.03)	0.394 (2.15)	0.312 (1.70)	0.244 (1.33)	0.157 (0.86)	0.280 (1.53)	0.243 (1.33)	0.183
d=50	bcf	0.274 (1.05)	0.586 (2.25)	0.480 (1.84)	0.238 (0.91)	0.169 (0.65)	0.327 (1.25)	0.352 (1.35)	0.260
effect_fn=additive	bcf	0.283 (1.09)	0.463 (1.78)	0.418 (1.61)	0.259 (1.00)	0.235 (0.90)	0.316 (1.22)	0.336 (1.29)	0.260
effect_fn=interact	bcf	0.297 (0.76)	0.496 (1.26)	0.466 (1.19)	0.278 (0.71)	0.275 (0.70)	0.418 (1.06)	0.411 (1.05)	0.392
effect_fn=rff	bcf	0.396 (0.93)	0.488 (1.15)	0.471 (1.11)	0.393 (0.92)	0.389 (0.91)	0.468 (1.10)	0.430 (1.01)	0.425
frac_det=0.05	bcf	0.217 (1.07)	0.446 (2.21)	0.363 (1.80)	0.233 (1.16)	0.171 (0.85)	0.280 (1.39)	0.281 (1.39)	0.202
frac_det=0.1	bcf	0.226 (1.09)	0.446 (2.14)	0.366 (1.76)	0.240 (1.16)	0.187 (0.90)	0.282 (1.35)	0.291 (1.40)	0.208
frac_det=0.2	bcf	0.246 (1.07)	0.450 (1.96)	0.370 (1.61)	0.311 (1.36)	0.274 (1.19)	0.308 (1.34)	0.309 (1.35)	0.229
gamma_u=0.5	cforest	0.622 (1.03)	0.571 (0.95)	0.568 (0.94)	0.657 (1.09)	0.597 (0.99)	0.692 (1.15)	0.601	0.619 (1.03)
gamma_u=1.0	cforest	0.962 (1.02)	0.795 (0.84)	0.796 (0.84)	0.962 (1.02)	0.950 (1.01)	0.989 (1.05)	0.942	0.962 (1.02)
gamma_u=2.0	cforest	1.316 (1.01)	1.036 (0.80)	1.058 (0.81)	1.315 (1.01)	1.312 (1.01)	1.337 (1.03)	1.300	1.322 (1.02)
het=0.0	cforest	0.167 (2.73)	0.136 (2.22)	0.135 (2.21)	0.228 (3.72)	0.108 (1.77)	0.254 (4.16)	0.061	0.078 (1.27)

cell	best cl.	CausalPFN	Do-PFN	CausalFM	TabPFN-T	TabPFN-DR	DR	CF	BCF
het=0.25	bcf	0.192 (1.38)	0.253 (1.82)	0.221 (1.59)	0.228 (1.64)	0.150 (1.08)	0.262 (1.89)	0.171 (1.23)	0.139
het=1.0	bcf	0.245 (0.86)	0.854 (3.00)	0.698 (2.45)	0.237 (0.83)	0.181 (0.64)	0.370 (1.30)	0.472 (1.66)	0.284
het=2.0	bcf	0.270 (0.67)	1.695 (4.22)	1.425 (3.55)	0.240 (0.60)	0.197 (0.49)	0.553 (1.38)	0.852 (2.12)	0.401
kappa=0.25	bcf	0.177 (0.99)	0.440 (2.46)	0.342 (1.91)	0.205 (1.14)	0.112 (0.63)	0.243 (1.36)	0.265 (1.48)	0.179
kappa=0.5	bcf	0.190 (1.01)	0.439 (2.33)	0.350 (1.86)	0.213 (1.13)	0.126 (0.67)	0.260 (1.38)	0.267 (1.42)	0.188
kappa=2.0	bcf	0.273 (1.20)	0.469 (2.07)	0.417 (1.84)	0.281 (1.24)	0.225 (0.99)	0.352 (1.56)	0.311 (1.37)	0.226
kappa=3.0	bcf	0.322 (1.35)	0.482 (2.01)	0.459 (1.92)	0.332 (1.38)	0.294 (1.23)	0.476 (1.99)	0.327 (1.36)	0.240
n=2000	bcf	0.182 (1.09)	0.433 (2.58)	0.359 (2.14)	0.196 (1.17)	0.134 (0.80)	0.226 (1.35)	0.239 (1.43)	0.168
n=250	bcf	0.353 (0.98)	0.474 (1.31)	0.381 (1.06)	0.411 (1.14)	0.333 (0.92)	0.588 (1.63)	0.408 (1.13)	0.361
n=4000	bcf	0.150 (1.09)	0.431 (3.14)	0.353 (2.57)	0.144 (1.05)	0.095 (0.69)	0.178 (1.30)	0.206 (1.50)	0.137
n=500	bcf	0.283 (1.05)	0.456 (1.68)	0.371 (1.37)	0.326 (1.20)	0.244 (0.90)	0.424 (1.56)	0.337 (1.24)	0.271
n=8000	bcf	0.143 (1.26)	0.432 (3.82)	0.353 (3.12)	0.116 (1.03)	0.076 (0.67)	0.163 (1.44)	0.179 (1.58)	0.113
noise=hetero	bcf	0.205 (1.05)	0.451 (2.30)	0.367 (1.88)	0.238 (1.21)	0.172 (0.88)	0.298 (1.52)	0.268 (1.37)	0.196
noise=laplace	bcf	0.211 (1.03)	0.445 (2.17)	0.362 (1.77)	0.240 (1.17)	0.158 (0.77)	0.290 (1.41)	0.281 (1.37)	0.205
noise_sd=0.1	bcf	0.129 (1.17)	0.437 (3.94)	0.368 (3.32)	0.120 (1.08)	0.078 (0.70)	0.206 (1.86)	0.221 (1.99)	0.111
noise_sd=1.0	bcf	0.289 (0.98)	0.448 (1.52)	0.376 (1.27)	0.355 (1.20)	0.257 (0.87)	0.445 (1.51)	0.337 (1.14)	0.295
noise_sd=2.0	bcf	0.418 (1.02)	0.461 (1.12)	0.412 (1.00)	0.553 (1.34)	0.416 (1.01)	0.772 (1.88)	0.416 (1.01)	0.411
outcome_fn=interact	bcf	0.184 (0.77)	0.482 (2.03)	0.368 (1.55)	0.264 (1.11)	0.168 (0.71)	0.333 (1.41)	0.299 (1.26)	0.237
outcome_fn=linear	bcf	0.120 (0.58)	0.456 (2.21)	0.363 (1.76)	0.113 (0.55)	0.129 (0.63)	0.254 (1.23)	0.292 (1.41)	0.206
outcome_fn=rff	cforest	0.369 (0.88)	0.428 (1.02)	0.434 (1.04)	0.481 (1.15)	0.373 (0.89)	0.489 (1.17)	0.418	0.431 (1.03)
rho_c=0.0	bcf	0.205 (1.03)	0.464 (2.33)	0.371 (1.87)	0.226 (1.14)	0.150 (0.75)	0.264 (1.33)	0.280 (1.41)	0.199
rho_c=0.3	bcf	0.204 (1.04)	0.456 (2.31)	0.363 (1.84)	0.225 (1.14)	0.150 (0.76)	0.279 (1.41)	0.277 (1.41)	0.197
rho_c=0.95	bcf	0.239 (1.14)	0.441 (2.10)	0.406 (1.94)	0.244 (1.16)	0.169 (0.81)	0.318 (1.52)	0.282 (1.34)	0.210
scale=0.01	bcf	0.002 (1.08)	0.007 (3.61)	0.004 (1.86)	0.002 (1.17)	0.002 (0.82)	0.003 (1.45)	0.003 (1.40)	0.002
scale=100.0	bcf	21.391 (1.08)	44.493 (2.25)	36.777 (1.86)	23.076 (1.17)	16.252 (0.82)	28.352 (1.44)	27.684 (1.40)	19.756
shift=100.0	bcf	0.214 (1.08)	0.445 (2.25)	0.368 (1.86)	0.231 (1.17)	0.163 (0.82)	0.286 (1.45)	0.277 (1.40)	0.198

Table 6. Development platform, coverage / mean width of nominal 90% CATE intervals on test units. src: `runs/e02_platform/analysis/d3_calibration.md`.

cell	CausalPFN	CausalPFN-cal	CausalFM	CF	BCF
base	1.00 / 3.64	1.00 / 2.64	0.93 / 1.41	0.55 / 0.39	0.87 / 0.57
ate=0.0	1.00 / 3.64	1.00 / 2.64	0.93 / 1.46	0.55 / 0.39	0.86 / 0.56
ate=0.25	1.00 / 3.64	1.00 / 2.64	0.93 / 1.42	0.55 / 0.38	0.86 / 0.56
ate=1.0	1.00 / 3.64	1.00 / 2.64	0.94 / 1.44	0.55 / 0.39	0.88 / 0.59
ate=2.0	1.00 / 3.64	1.00 / 2.64	0.94 / 1.60	0.56 / 0.40	0.90 / 0.62
ate=4.0	1.00 / 3.64	1.00 / 2.64	0.89 / 2.32	0.55 / 0.42	0.91 / 0.68
covariates=binary	1.00 / 3.47	1.00 / 2.42	0.98 / 1.81	0.75 / 0.43	0.89 / 0.36
covariates=corr	1.00 / 3.56	1.00 / 2.58	0.93 / 1.45	0.56 / 0.38	0.84 / 0.56
covariates=mixed	1.00 / 3.57	1.00 / 2.54	0.95 / 1.78	0.59 / 0.40	0.87 / 0.50
d=100	1.00 / 4.58	1.00 / 3.75	0.97 / 2.38	0.39 / 0.34	0.77 / 0.62
d=150	1.00 / 4.92	1.00 / 4.27	0.97 / 2.51	0.34 / 0.33	0.73 / 0.63
d=25	1.00 / 3.66	1.00 / 2.87	0.91 / 1.61	0.46 / 0.36	0.82 / 0.60
d=5	1.00 / 3.63	1.00 / 2.57	0.98 / 1.73	0.65 / 0.41	0.87 / 0.52
d=50	1.00 / 3.75	1.00 / 2.88	0.95 / 2.02	0.40 / 0.35	0.78 / 0.61
effect_fn=additive	1.00 / 3.64	1.00 / 2.69	0.89 / 1.36	0.48 / 0.38	0.80 / 0.57
effect_fn=interact	1.00 / 3.63	1.00 / 2.71	0.88 / 1.39	0.44 / 0.35	0.66 / 0.55
effect_fn=rff	1.00 / 3.79	1.00 / 2.82	0.85 / 1.40	0.31 / 0.34	0.50 / 0.55
frac_det=0.05	1.00 / 3.66	1.00 / 2.65	0.93 / 1.42	0.54 / 0.39	0.86 / 0.57
frac_det=0.1	1.00 / 3.66	1.00 / 2.63	0.93 / 1.43	0.53 / 0.39	0.85 / 0.57
frac_det=0.2	1.00 / 3.66	1.00 / 2.60	0.93 / 1.46	0.52 / 0.40	0.81 / 0.57
gamma_u=0.5	1.00 / 6.16	1.00 / 5.49	0.94 / 2.26	0.28 / 0.61	0.33 / 0.82
gamma_u=1.0	1.00 / 5.89	1.00 / 5.29	0.82 / 2.18	0.05 / 0.59	0.05 / 0.82
gamma_u=2.0	1.00 / 5.44	1.00 / 4.80	0.52 / 2.07	0.00 / 0.53	0.00 / 0.80
het=0.0	1.00 / 3.52	1.00 / 2.60	1.00 / 1.28	0.94 / 0.25	0.99 / 0.41
het=0.25	1.00 / 3.55	1.00 / 2.60	0.99 / 1.31	0.62 / 0.30	0.92 / 0.49
het=1.0	1.00 / 3.93	1.00 / 2.72	0.81 / 1.80	0.51 / 0.56	0.83 / 0.69
het=2.0	1.00 / 4.84	1.00 / 2.98	0.74 / 3.00	0.46 / 0.89	0.81 / 0.86
kappa=0.25	1.00 / 3.69	1.00 / 2.62	0.96 / 1.45	0.56 / 0.37	0.90 / 0.56
kappa=0.5	1.00 / 3.67	1.00 / 2.63	0.95 / 1.44	0.57 / 0.38	0.88 / 0.56
kappa=2.0	1.00 / 3.60	1.00 / 2.65	0.89 / 1.41	0.53 / 0.41	0.82 / 0.57
kappa=3.0	1.00 / 3.57	1.00 / 2.65	0.85 / 1.41	0.52 / 0.43	0.81 / 0.59
n=2000	1.00 / 3.61	1.00 / 2.56	0.94 / 1.44	0.59 / 0.35	0.87 / 0.48
n=250	1.00 / 3.81	1.00 / 3.26	0.93 / 1.54	0.47 / 0.50	0.72 / 0.75
n=4000	1.00 / 3.48	1.00 / 2.43	0.94 / 1.44	0.62 / 0.31	0.85 / 0.37
n=500	1.00 / 3.69	1.00 / 2.90	0.93 / 1.46	0.51 / 0.43	0.81 / 0.66
n=8000	1.00 / 3.39	1.00 / 2.36	0.94 / 1.41	0.64 / 0.29	0.83 / 0.30
noise=hetero	1.00 / 3.52	1.00 / 2.20	0.94 / 1.43	0.55 / 0.38	0.87 / 0.56
noise=laplace	1.00 / 3.27	1.00 / 2.22	0.94 / 1.48	0.53 / 0.38	0.85 / 0.56
noise_sd=0.1	1.00 / 2.56	0.99 / 0.80	0.87 / 1.16	0.49 / 0.25	0.76 / 0.24
noise_sd=1.0	1.00 / 5.73	1.00 / 5.01	0.99 / 2.12	0.61 / 0.57	0.83 / 0.78
noise_sd=2.0	1.00 / 10.40	1.00 / 9.73	1.00 / 3.97	0.75 / 0.99	0.83 / 1.15
outcome_fn=interact	1.00 / 3.56	1.00 / 2.55	0.90 / 1.32	0.52 / 0.41	0.81 / 0.60
outcome_fn=linear	1.00 / 3.42	1.00 / 2.38	0.87 / 1.14	0.50 / 0.38	0.85 / 0.56
outcome_fn=rff	1.00 / 4.40	1.00 / 3.70	0.96 / 1.89	0.42 / 0.49	0.52 / 0.66
rho_c=0.0	1.00 / 3.69	1.00 / 2.65	0.94 / 1.44	0.55 / 0.38	0.86 / 0.56

cell	CausalPFN	CausalPFN-cal	CausalFM	CF	BCF
rho_c=0.3	1.00 / 3.68	1.00 / 2.65	0.94 / 1.43	0.55 / 0.38	0.87 / 0.57
rho_c=0.95	1.00 / 3.58	1.00 / 2.63	0.89 / 1.37	0.54 / 0.39	0.85 / 0.57
scale=0.01	1.00 / 0.04	1.00 / 0.03	0.93 / 0.01	0.55 / 0.00	0.87 / 0.01
scale=100.0	1.00 / 364.03	1.00 / 263.58	0.93 / 140.97	0.55 / 38.64	0.87 / 56.79
shift=100.0	1.00 / 3.64	1.00 / 2.64	0.93 / 1.41	0.55 / 0.39	0.87 / 0.57

Table 7. Development platform, coverage (share of 20 seeds) / mean width of nominal 95% ATE intervals. src: `runs/e02_platform/analysis/d3_calibration.md`.

cell	CausalPFN	CausalPFN-cal	TabPFN-DR	DR	CF	BCF
base	0.55 / 0.137	0.35 / 0.098	0.95 / 0.188	0.80 / 0.180	1.00 / 0.501	0.85 / 0.146
ate=0.0	0.65 / 0.137	0.35 / 0.098	0.95 / 0.188	0.80 / 0.180	1.00 / 0.499	0.90 / 0.147
ate=0.25	0.60 / 0.136	0.40 / 0.098	0.95 / 0.188	0.80 / 0.180	1.00 / 0.498	0.90 / 0.147
ate=1.0	0.55 / 0.134	0.35 / 0.099	0.95 / 0.188	0.80 / 0.179	1.00 / 0.506	0.80 / 0.152
ate=2.0	0.55 / 0.136	0.35 / 0.098	0.95 / 0.188	0.80 / 0.180	1.00 / 0.519	0.85 / 0.151
ate=4.0	0.60 / 0.136	0.35 / 0.098	0.95 / 0.188	0.80 / 0.180	1.00 / 0.543	0.95 / 0.159
covariates=binary	0.75 / 0.129	0.65 / 0.091	0.95 / 0.167	0.95 / 0.160	1.00 / 0.532	0.85 / 0.143
covariates=corr	0.70 / 0.133	0.50 / 0.096	0.95 / 0.203	0.90 / 0.189	1.00 / 0.499	0.85 / 0.151
covariates=mixed	0.85 / 0.135	0.55 / 0.094	0.90 / 0.182	0.85 / 0.173	1.00 / 0.510	0.85 / 0.147
d=100	0.10 / 0.169	0.05 / 0.139	1.00 / 0.182	0.35 / 0.183	1.00 / 0.440	0.75 / 0.151
d=150	0.10 / 0.185	0.10 / 0.161	0.95 / 0.178	0.35 / 0.186	1.00 / 0.427	0.75 / 0.150
d=25	0.55 / 0.135	0.30 / 0.106	0.95 / 0.206	0.55 / 0.182	1.00 / 0.465	0.55 / 0.150
d=5	0.70 / 0.137	0.60 / 0.097	1.00 / 0.194	0.95 / 0.181	1.00 / 0.525	0.95 / 0.148
d=50	0.55 / 0.137	0.35 / 0.106	0.85 / 0.176	0.45 / 0.180	1.00 / 0.444	0.60 / 0.152
effect_fn=additive	0.65 / 0.136	0.55 / 0.101	0.95 / 0.219	0.75 / 0.181	1.00 / 0.495	0.75 / 0.151
effect_fn=interact	0.65 / 0.136	0.60 / 0.101	0.95 / 0.215	0.80 / 0.186	1.00 / 0.450	0.80 / 0.158
effect_fn=rff	0.65 / 0.141	0.50 / 0.105	1.00 / 0.199	0.85 / 0.191	1.00 / 0.442	0.70 / 0.163
frac_det=0.05	0.60 / 0.137	0.45 / 0.100	0.90 / 0.196	0.70 / 0.173	1.00 / 0.504	0.90 / 0.145
frac_det=0.1	0.55 / 0.138	0.50 / 0.097	0.70 / 0.201	0.70 / 0.170	1.00 / 0.504	0.85 / 0.152
frac_det=0.2	0.55 / 0.137	0.40 / 0.097	0.60 / 0.168	0.55 / 0.164	1.00 / 0.521	0.70 / 0.172
gamma_u=0.5	0.00 / 0.231	0.00 / 0.206	0.00 / 0.376	0.00 / 0.345	0.20 / 0.786	0.00 / 0.312
gamma_u=1.0	0.00 / 0.222	0.00 / 0.199	0.00 / 0.346	0.00 / 0.328	0.00 / 0.758	0.00 / 0.292
gamma_u=2.0	0.00 / 0.204	0.00 / 0.182	0.00 / 0.275	0.00 / 0.291	0.00 / 0.697	0.00 / 0.261
het=0.0	0.70 / 0.132	0.45 / 0.097	0.85 / 0.176	0.75 / 0.166	1.00 / 0.330	0.85 / 0.149
het=0.25	0.60 / 0.133	0.40 / 0.097	0.85 / 0.179	0.80 / 0.170	1.00 / 0.386	0.85 / 0.148
het=1.0	0.85 / 0.147	0.40 / 0.102	1.00 / 0.219	0.90 / 0.213	1.00 / 0.719	0.90 / 0.152
het=2.0	0.85 / 0.183	0.55 / 0.113	1.00 / 0.310	1.00 / 0.310	1.00 / 1.130	0.90 / 0.150
kappa=0.25	0.80 / 0.138	0.70 / 0.098	0.95 / 0.150	0.95 / 0.171	1.00 / 0.480	0.95 / 0.134
kappa=0.5	0.80 / 0.137	0.65 / 0.098	0.95 / 0.163	1.00 / 0.176	1.00 / 0.488	0.95 / 0.137
kappa=2.0	0.20 / 0.134	0.20 / 0.100	0.90 / 0.297	0.55 / 0.184	1.00 / 0.527	0.80 / 0.170
kappa=3.0	0.25 / 0.134	0.25 / 0.099	0.70 / 0.339	0.70 / 0.296	1.00 / 0.560	0.75 / 0.190
n=2000	0.40 / 0.095	0.40 / 0.068	0.95 / 0.132	0.65 / 0.118	1.00 / 0.454	0.70 / 0.106
n=250	0.75 / 0.277	0.55 / 0.237	0.80 / 0.430	0.90 / 0.491	0.95 / 0.624	0.80 / 0.329
n=4000	0.55 / 0.065	0.45 / 0.046	1.00 / 0.091	0.85 / 0.078	1.00 / 0.406	0.90 / 0.071
n=500	0.55 / 0.194	0.35 / 0.153	0.85 / 0.265	0.85 / 0.301	1.00 / 0.543	0.80 / 0.221
n=8000	0.50 / 0.045	0.30 / 0.031	1.00 / 0.062	1.00 / 0.060	1.00 / 0.376	0.95 / 0.050
noise=hetero	0.55 / 0.134	0.40 / 0.084	0.95 / 0.202	0.85 / 0.182	1.00 / 0.506	0.90 / 0.145
noise=laplace	0.60 / 0.121	0.45 / 0.083	1.00 / 0.197	0.80 / 0.180	1.00 / 0.490	0.85 / 0.151
noise_sd=0.1	0.80 / 0.096	0.25 / 0.030	0.90 / 0.082	0.75 / 0.104	1.00 / 0.325	1.00 / 0.042
noise_sd=1.0	0.70 / 0.216	0.65 / 0.188	0.95 / 0.342	0.90 / 0.312	1.00 / 0.742	0.85 / 0.284
noise_sd=2.0	0.60 / 0.392	0.60 / 0.363	0.95 / 0.652	0.90 / 0.580	1.00 / 1.273	0.85 / 0.537
outcome_fn=interact	0.75 / 0.134	0.65 / 0.096	0.90 / 0.183	0.55 / 0.204	1.00 / 0.531	0.65 / 0.160
outcome_fn=linear	0.80 / 0.127	0.65 / 0.089	0.95 / 0.168	0.85 / 0.167	1.00 / 0.482	0.80 / 0.144
outcome_fn=rff	0.15 / 0.163	0.10 / 0.138	0.25 / 0.255	0.15 / 0.254	0.65 / 0.640	0.15 / 0.232
rho_c=0.0	0.95 / 0.138	0.90 / 0.099	1.00 / 0.195	1.00 / 0.178	1.00 / 0.491	0.95 / 0.147
rho_c=0.3	0.75 / 0.138	0.75 / 0.098	1.00 / 0.186	1.00 / 0.181	1.00 / 0.499	1.00 / 0.146
rho_c=0.95	0.35 / 0.135	0.20 / 0.098	0.80 / 0.181	0.65 / 0.184	1.00 / 0.497	0.60 / 0.151
scale=0.01	0.55 / 0.001	0.35 / 0.001	0.95 / 0.002	0.80 / 0.002	1.00 / 0.005	0.85 / 0.001
scale=100.0	0.55 / 13.695	0.35 / 9.764	0.95 / 18.823	0.80 / 17.964	1.00 / 50.112	0.85 / 14.554
shift=100.0	0.55 / 0.136	0.35 / 0.098	0.95 / 0.188	0.80 / 0.180	1.00 / 0.500	0.85 / 0.146

D. Proofs

Throughout, P denotes expectation over a new observation $O = (X, T, Y)$, $\mathbb{P}_{n,k}$ the empirical measure on fold I_k ($|I_k| = n/K$), and $\hat{\eta}_k$, g_k, r_k the nuisance estimates, PFN estimate and reference estimate obtained from $\mathcal{D}_{-k}$. For the proofs we strengthen Assumption A2 to bounded nuisance estimates, $|\hat{m}_{tk}| \le M$, and add a stability condition, standard for cross-validated risk estimates.

Assumption A6 (Stability)**.** There are fixed functions g^*, r^* with $\max_k \left(\|g_k - g^*\|_2 + \|r_k - r^*\|_2 + \|\hat{\eta}_k - \eta\|_2 \right) = o_p(1)$.

Assumption A6 does not require the PFN to be consistent for τ, only that its output on $(1 - 1/K)n$ units settles as n grows.

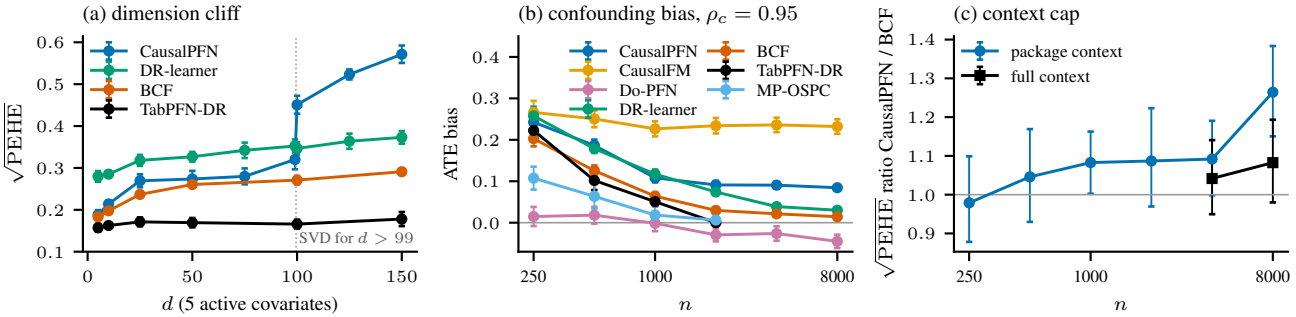


Figure 3. **Silent limits and a model-level bias.** (a) $\sqrt{\text{PEHE}}$ vs. d (5 active covariates): CausalPFN jumps at $d = 100$, where its wrapper switches on an unsupervised SVD. (b) ATE bias under strong confounding vs. n: CausalPFN and CausalFM plateau; the orthogonal estimators do not. (c) CausalPFN/BCF $\sqrt{\text{PEHE}}$ ratio vs. n with the package's context cap and with full context. Bars: (a, b) ±1 s.e. over seeds (20 in a, 30 in b); (c) 95% paired-bootstrap intervals over 20 seeds.

Proof of Proposition 5.1. Write $\delta_k(O) = (g_k - r_k)(X)\,[(g_k + r_k)(X) - 2\varphi(O; \hat{\eta}_k)]$ and $\delta^*(O) = (g^* - r^*)(X)\,[(g^* + r^*)(X) - 2\varphi(O; \eta)]$. Then

$$\hat{\Delta} = \frac{1}{K}\sum_k \mathbb{P}_{n,k}\delta_k = (\mathbb{P}_n - P)\delta^* + \frac{1}{K}\sum_k (\mathbb{P}_{n,k} - P)(\delta_k - \delta^*) + \frac{1}{K}\sum_k P\delta_k.$$

Second term. Conditionally on $\mathcal{D}_{-k}$, $\delta_k - \delta^*$ is a fixed function, and the units in I_k are i.i.d., so $\mathbb{E}[(\mathbb{P}_{n,k} - P)(\delta_k - \delta^*) \mid \mathcal{D}_{-k}] = 0$ and its conditional variance is at most $\frac{K}{n}\|\delta_k - \delta^*\|_2^2$. Under Assumptions A1 and A2 with clipping, δ is a Lipschitz function of (g, r, φ) with constants depending on M and c, and φ is Lipschitz in $\hat{\eta}$; by Assumption A6, $\|\delta_k - \delta^*\|_2 = o_p(1)$. Chebyshev's inequality conditionally on $\mathcal{D}_{-k}$ gives $(\mathbb{P}_{n,k} - P)(\delta_k - \delta^*) = o_p(n^{-1/2})$, the standard sample-splitting argument (Chernozhukov et al., 2018; Kennedy, 2022). *Third term.* Since $\mathbb{E}[\varphi(O; \hat{\eta}_k) \mid X, \mathcal{D}_{-k}] = \tau(X) + b_k(X)$,

$$P\delta_k = \mathbb{E}_X\big[(g_k - r_k)(g_k + r_k - 2\tau)\big] - 2\,\mathbb{E}_X[(g_k - r_k)\,b_k] = \Delta_k - 2\,\mathbb{E}_X[(g_k - r_k)\,b_k],$$

because $(g-r)(g+r-2\tau) = (g-\tau)^2 - (r-\tau)^2$. For the AIPW score, $b_k = (e - \hat{e}_k)\,[(\mu_1 - \hat{m}_{1k})/\hat{e}_k + (\mu_0 - \hat{m}_{0k})/(1-\hat{e}_k)]$, so by Cauchy–Schwarz and clipping, $|\mathbb{E}_X[(g_k - r_k)\,b_k]| \le \frac{2M}{c}\,\|e - \hat{e}_k\|_2\,\big(\|\mu_1 - \hat{m}_{1k}\|_2 + \|\mu_0 - \hat{m}_{0k}\|_2\big) = o_p(n^{-1/2})$ by Assumption A3. Hence $\hat{\Delta} - \bar{\Delta} = (\mathbb{P}_n - P)\delta^* + o_p(n^{-1/2})$. $\delta^*(O_i)$ are i.i.d., bounded, with variance $\sigma_*^2 \ge \sigma_0^2 > 0$ (Assumption A4, passed to the limit), so the central limit theorem gives $\sqrt{n}(\hat{\Delta} - \bar{\Delta}) \Rightarrow \mathcal{N}(0, \sigma_*^2)$. The same decomposition applied to δ^2 shows $\hat{\sigma}^2 \to \sigma_*^2$ in probability, and Slutsky's lemma completes the proof. □

Proof of Proposition 5.2. *Regional statistics.* Let $G_k(x) = 1\{\hat{e}_k(x) \in [0.1, 0.9]\}$ for units of fold k (and its complement for G_{out}). Assume $\mathbb{P}(e(X) \in \{0.1, 0.9\}) = 0$, so $\|G_k - G^*\|_2 = o_p(1)$ with $G^*(x) = 1\{e(x) \in [0.1, 0.9]\}$ by Assumption A6. The regional mean is the ratio $\hat{\Delta}_g = \mathbb{P}_n(\delta\, G)/\mathbb{P}_n(G)$; applying the argument of Proposition 5.1 to numerator and denominator (the indicator is a bounded fixed function given $\mathcal{D}_{-k}$) and the delta method gives $\sqrt{n_g}(\hat{\Delta}_g - \bar{\Delta}_g)/\hat{\sigma}_g \Rightarrow \mathcal{N}(0, 1)$ for each of the finitely many (g, r), provided the region has positive mass. *(i) Size.* If $\bar{\Delta}_{g,r} \le 0$, the one-sided p-value $p_{g,r} = 1 - \Phi(Z_{g,r})$ satisfies $\limsup \mathbb{P}(p_{g,r} \le u) \le u$ for every u. Holm's procedure controls the family-wise error rate at level α under arbitrary dependence among valid p-values, so, with finitely many tests, the probability of rejecting any hypothesis is at most $\alpha + o(1)$. Deferral happens only after a rejection. *(ii) Power.* If $\bar{\Delta}_{g,r} \ge \delta_0 > 0$, then $Z_{g,r} \to \infty$ in probability, so $p_{g,r} \to 0$; Holm rejects every hypothesis whose p-value is below α/m (m the number of tests), so $H_0^{g,r}$ is rejected and the procedure defers in region g with probability $\to 1$. *(iii) Risk.* Let F be the event that some true null is rejected; $\mathbb{P}(F) \le \alpha + o(1)$ by (i) applied to the true nulls (Holm controls the family-wise error rate in the strong sense). On F^c, every deferral in region g goes to a reference with $\bar{\Delta}_{g,r} > 0$, hence, for $|\bar{\Delta}_{g,r}| \ge \epsilon_n$, with $R_{n,g}(r) < R_{n,g}(f)$ by Assumption A5 applied regionwise; references with $|\bar{\Delta}_{g,r}| < \epsilon_n$ change the risk by $o(1)$. On F, the excess risk in region g is at most $\max_r (R_{n,g}(r) - R_{n,g}(f))_+$. Since $R_n(\hat{h}) = \sum_g w_g R_{n,g}(\hat{h})$, taking expectations gives the bound; the $o(1)$ terms absorb the difference between the regions defined by $\hat{e}_k$ (statistics) and by $\hat{e}$ fitted on $\mathcal{D}$ (test units), which vanishes under Assumption A6 and the no-atom condition. If in region g a unique reference maximizes $\bar{\Delta}_{g,r} > 0$ with a positive margin, consistency of $\hat{\Delta}_{g,r}$ implies that the arg max over rejected references picks it with probability $\to 1$. □

E. Method Configuration and Variants

All of the following was fixed on development data only (pilot seeds 0–39). In-context cross-fitting uses $K = 5$ folds stratified on T (shuffled, seeded by the method seed); the final CATE of the PFN on the test units uses the full context, so always-PFN equals the plain PFN output (verified on every task, maximum absolute difference below 10^{-8}). References: the DR-learner and BCF of Appendix A, cross-fitted on the same folds and refit on all data for the test units. Scoring nuisances: $\hat{e}$, $\hat{m}_0$, $\hat{m}_1$, each the average of LightGBM (300 trees, learning rate 0.03, 15 leaves, minimum child size 20, row and column subsampling 0.8) and a random forest (200 trees, minimum leaf 5, 50% of features per split), fit out of fold; $\hat{e}$ clipped to $[0.01, 0.99]$. Reference fits and scoring nuisances do not depend on the PFN; they are cached per task (content hash of the public data) and shared between PFNs, and their wall time is charged to every PFN that uses them. Do-PFN is evaluated only as a plain baseline: it needs a separate software environment, and wrapping it was not part of the protocol.

Algorithm 1 gives the procedure as evaluated.

Algorithm 1 Certify-or-defer (region-wise, as evaluated: A-V5)

Input: data $\mathcal{D}$, frozen PFN f, references $\mathcal{R}$, test points X^*; $K = 5$, $c = 0.01$, $\alpha = 0.05$
Split $\mathcal{D}$ into K folds stratified on T
for $k = 1$ **to** K **do**
  $g(X_{I_k}) \leftarrow f(\mathcal{D}_{-k}, X_{I_k})$ {one forward pass}
  fit $r \in \mathcal{R}$ and $\hat{\eta} = (\hat{e}, \hat{m}_0, \hat{m}_1)$ on $\mathcal{D}_{-k}$; predict I_k
end for
compute φ_i and $\delta_i^{(r)}$ for all i, r (Equation (3))
for $g \in \{\text{in}, \text{out}\}$, $r \in \mathcal{R}$ **do**
  $Z_{g,r} \leftarrow \sqrt{|G_g|}\,\text{mean}_{i \in G_g} \delta_i^{(r)} / \text{sd}_{i \in G_g} \delta_i^{(r)}$
end for
reject by Holm at level α over all one-sided tests
$\hat{h}(X^*) \leftarrow f(\mathcal{D}, X^*)$ {full context, as the base PFN}
for g with a rejected test **do**
  $r^\star \leftarrow \arg\max_{\text{rejected } r} \hat{\Delta}_{g,r}$; $\hat{h}(X^*_{G_g}) \leftarrow r^\star(X^*_{G_g})$ {$r^\star$ fit on $\mathcal{D}$}
end for
Output: $\hat{h}$; certificate (Z_{DR}, overlap share, BLP slope)

The pilots evaluated five variants from the same saved per-unit arrays: V1, the single-region test with $\mathcal{R} = \{\text{DR, BCF}\}$; V2, $\mathcal{R} = \{\text{DR}\}$; V3, $\mathcal{R}$ extended by causal isotonic calibration of the PFN (van der Laan et al., 2023) with nested out-of-fold predictions; V4, clipping at 0.05; V5, the region-wise test of Section 5 (A-V5). For CATE-only models the ATE estimate uses $\hat{\mu}_1 = \hat{m} + (1 - \hat{e})\hat{\tau}_f$ and $\hat{\mu}_0 = \hat{m} - \hat{e}\,\hat{\tau}_f$ with $\hat{m} = \hat{e}\hat{m}_1 + (1 - \hat{e})\hat{m}_0$, from the out-of-fold scoring nuisances.

F. Pilot Results

Table 8. Pre-registered pilot outcomes for the region-wise variant (A-V5). P01: seeds 0–19 (its registered output was the single-region variant); P02: fresh seeds 20–39, A-V5 registered. Gain: share of oracle gain recovered over failure cells; harm: share of deferred tasks made worse.

		CausalPFN	CausalFM
P-1	P01	0.946 vs. 0.884 ✓	0.888 vs. 0.652 ✓
	P02	0.978 vs. 0.958 ×	0.901 vs. 0.626 ✓
P-2	P01	gain 0.50, harm 0.09 ×	gain 0.72, harm 0.09 ✓
	P02	gain 0.62, harm 0.09 ✓	gain 0.72, harm 0.05 ✓
P-3	P01	× (5 cells < 0.90)	× (6 cells < 0.90)

Harmful deferrals of A-V5 in P02: CausalPFN 3 of 35 deferring tasks (hard overlap $+12\%$ and $+27\%$, $n = 4000$ $+28\%$ in $\sqrt{\text{PEHE}}$ over always-PFN); CausalFM 7 of 138 (median $+12\%$; hard overlap 2, hidden confounding, effect s.d. 0, $\kappa = 3$, random-feature outcome 2). In P01, the single-region variant was the registered output; the tables below report every variant's output computed from the same arrays.

Table 9. Pilot P01, CausalPFN: $\sqrt{\text{PEHE}}$ per cell (mean over 20 seeds) of always-PFN, the single-region and region-wise (A-V5) outputs, the references, causal isotonic calibration and the oracle; share of tasks with the failure event; coverage (width) of A's AIPW ATE interval. src: `p01_certify/analysis/p01.md`.

cell	k	always-PFN	single-region	A-V5	DR	BCF	iso-cal	oracle	fail share	AIPW ATE cov (width)
base	20	0.214	0.214	0.214	0.285	0.198	0.299	0.189	0.00	0.95 (0.218)
n=4000	20	0.150	0.151	0.151	0.178	0.137	0.174	0.131	0.20	1.00 (0.087)
het=2.0	20	0.270	0.270	0.270	0.553	0.401	0.524	0.270	0.00	1.00 (0.330)
n=250	20	0.353	0.349	0.353	0.588	0.361	0.527	0.320	0.00	0.95 (0.515)
d=100	20	0.451	0.348	0.366	0.347	0.271	0.428	0.269	0.80	0.55 (0.259)
d=150	20	0.571	0.342	0.340	0.373	0.291	0.487	0.291	0.95	0.45 (0.304)
het=0.0	20	0.167	0.165	0.165	0.254	0.078	0.149	0.078	0.00	0.90 (0.202)
kappa=3.0	20	0.322	0.322	0.322	0.476	0.240	0.433	0.235	0.00	0.70 (0.252)
frac_det=0.1	20	0.226	0.227	0.225	0.282	0.208	0.305	0.197	0.00	0.80 (0.201)
outcome_fn=rff	20	0.369	0.369	0.369	0.489	0.431	0.438	0.369	0.00	0.35 (0.266)
effect_fn=rff	20	0.396	0.396	0.396	0.468	0.425	0.432	0.391	0.00	1.00 (0.224)
gamma_u=1.0	20	0.962	0.962	0.962	0.989	0.962	0.934	0.941	0.00	0.00 (0.385)

Table 10. Pilot P01, CausalFM: $\sqrt{\text{PEHE}}$ per cell (mean over 20 seeds) of always-PFN, the single-region and region-wise (A-V5) outputs, the references, causal isotonic calibration and the oracle; share of tasks with the failure event; coverage (width) of A's AIPW ATE interval. src: `p01_certify/analysis/p01.md`.

cell	k	always-PFN	single-region	A-V5	DR	BCF	iso-cal	oracle	fail share	AIPW ATE cov (width)
base	20	0.368	0.259	0.264	0.285	0.198	0.324	0.198	0.85	1.00 (0.248)
n=4000	20	0.353	0.152	0.159	0.178	0.137	0.230	0.137	1.00	1.00 (0.095)
het=2.0	20	1.425	0.401	0.510	0.553	0.401	0.833	0.401	1.00	1.00 (0.517)
n=250	20	0.381	0.375	0.381	0.588	0.361	0.520	0.354	0.00	0.85 (0.578)
d=100	20	0.496	0.301	0.304	0.347	0.271	0.483	0.269	0.90	0.60 (0.222)
d=150	20	0.510	0.330	0.329	0.373	0.291	0.500	0.291	0.95	0.50 (0.226)
het=0.0	20	0.135	0.129	0.130	0.254	0.078	0.092	0.078	0.05	1.00 (0.217)
kappa=3.0	20	0.459	0.407	0.416	0.476	0.240	0.458	0.240	0.30	0.60 (0.299)
frac_det=0.1	20	0.366	0.270	0.301	0.282	0.208	0.328	0.208	0.90	0.80 (0.222)
outcome_fn=rff	20	0.434	0.436	0.413	0.489	0.431	0.468	0.376	0.20	0.20 (0.305)
effect_fn=rff	20	0.471	0.462	0.444	0.468	0.425	0.482	0.421	0.40	0.90 (0.246)
gamma_u=1.0	20	0.796	0.832	0.821	0.989	0.962	0.946	0.796	0.00	0.00 (0.395)

Table 11. Pilot P02, CausalPFN: $\sqrt{\text{PEHE}}$ per cell (mean over 20 seeds) of always-PFN, the single-region and region-wise (A-V5) outputs, the references, causal isotonic calibration and the oracle; share of tasks with the failure event; coverage (width) of A's AIPW ATE interval. src: `p02_certify/analysis/p02.md`.

cell	k	always-PFN	single-region	A-V5	DR	BCF	iso-cal	oracle	fail share	AIPW ATE cov (width)
base	20	0.194	0.202	0.194	0.287	0.201	0.266	0.180	0.00	1.00 (0.212)
n=4000	20	0.129	0.131	0.131	0.184	0.135	0.172	0.120	0.00	0.90 (0.087)
het=2.0	20	0.257	0.257	0.257	0.559	0.400	0.449	0.256	0.00	1.00 (0.324)
n=250	20	0.315	0.315	0.315	0.548	0.349	0.500	0.303	0.00	1.00 (0.524)
d=100	20	0.473	0.323	0.342	0.336	0.264	0.453	0.264	0.85	0.65 (0.261)
d=150	20	0.580	0.326	0.325	0.361	0.285	0.502	0.281	1.00	0.45 (0.307)
het=0.0	20	0.150	0.153	0.148	0.251	0.085	0.167	0.085	0.00	1.00 (0.200)
kappa=3.0	20	0.274	0.274	0.274	0.456	0.230	0.421	0.216	0.00	0.70 (0.252)
frac_det=0.1	20	0.207	0.213	0.211	0.299	0.210	0.317	0.186	0.00	0.95 (0.196)
outcome_fn=rff	20	0.375	0.375	0.375	0.479	0.439	0.430	0.375	0.00	0.35 (0.274)
effect_fn=rff	20	0.410	0.410	0.410	0.460	0.430	0.440	0.407	0.00	0.95 (0.236)
gamma_u=1.0	20	0.951	0.951	0.951	1.007	0.955	0.935	0.940	0.00	0.00 (0.368)

Table 12. Pilot P02, CausalFM: $\sqrt{\text{PEHE}}$ per cell (mean over 20 seeds) of always-PFN, the single-region and region-wise (A-V5) outputs, the references, causal isotonic calibration and the oracle; share of tasks with the failure event; coverage (width) of A's AIPW ATE interval. src: `p02_certify/analysis/p02.md`.

cell	k	always-PFN	single-region	A-V5	DR	BCF	iso-cal	oracle	fail share	AIPW ATE cov (width)
base	20	0.367	0.257	0.270	0.287	0.201	0.326	0.201	0.85	0.95 (0.254)
n=4000	20	0.351	0.157	0.164	0.184	0.135	0.245	0.135	1.00	0.90 (0.096)
het=2.0	20	1.332	0.415	0.482	0.559	0.400	0.790	0.399	1.00	1.00 (0.493)
n=250	20	0.394	0.378	0.380	0.548	0.349	0.508	0.338	0.00	1.00 (0.641)
d=100	20	0.506	0.281	0.294	0.336	0.264	0.489	0.264	1.00	0.55 (0.229)
d=150	20	0.506	0.326	0.316	0.361	0.285	0.506	0.281	1.00	0.50 (0.235)
het=0.0	20	0.165	0.150	0.148	0.251	0.085	0.109	0.085	0.05	1.00 (0.226)
kappa=3.0	20	0.457	0.390	0.398	0.456	0.230	0.432	0.230	0.35	0.65 (0.310)
frac_det=0.1	20	0.372	0.275	0.298	0.299	0.210	0.356	0.209	0.80	0.85 (0.225)

cell	k	always-PFN	single-region	A-V5	DR	BCF	iso-cal	oracle	fail share	AIPW ATE cov (width)
outcome_fn=rff	20	0.453	0.430	0.440	0.479	0.439	0.444	0.403	0.35	0.25 (0.313)
effect_fn=rff	20	0.469	0.442	0.444	0.460	0.430	0.482	0.423	0.25	0.85 (0.261)
gamma_u=1.0	20	0.814	0.838	0.819	1.007	0.955	0.948	0.814	0.00	0.00 (0.388)

G. Final Evaluation: Full Results

Consistency checks, run before any claim was computed: record status (13,517 records, 0 not `ok`); code version (only the two frozen versions are accepted; found: 24626b8, 5b04464); A-V5's output recomputed offline from the saved arrays against the registered output; and always-PFN taken from A-V5's arrays against the plain PFN run of the same task. The offline recomputation reproduces the registered output exactly (maximum absolute difference 0 on every benchmark), and always-PFN matches the plain PFN run to within 8.0×10^{-7} on the noiseless benchmarks and 7.6×10^{-5} on B3 / B4, whose outcomes are in dollars. The 1,760 robustness records (two extra method seeds on B1 and H at $n = 500$) are all `ok` and on 5b04464. The first analysis run stopped before writing any output, on a field that only the development records carry; the fix (9231523) leaves the development dry-run output byte-identical. Table 22 is an exploratory follow-up written after the verdicts. Tables 13 to 18 give every method on every benchmark, followed by A-V5 against each method; Table 19 splits H by mechanism, Table 20 gives the cost and Table 21 the seed robustness.

Table 13. B1 IHDP: every method on the same tasks. $\sqrt{\text{PEHE}}$ and |ATE err| as mean ± s.e.; sanity: × if the mean |ATE err| is worse than the naive difference in means (0.286); CATE 90% and ATE 95% coverage (mean width); deferred: share of tasks where A-V5 deferred in some region.

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
CausalPFN	100	0.579 ± 0.070	0.097 ± 0.010	ok	1.00 (12.1)	0.89 (0.550)	
CausalPFN (temp.-cal.)	100	0.579 ± 0.070	0.097 ± 0.010	ok	1.00 (4.933)	0.62 (0.226)	
CausalFM	100	5.860 ± 0.728	2.352 ± 0.084	×	0.43 (10.1)	–	
Do-PFN	100	6.065 ± 0.899	1.910 ± 0.256	×	–	–	
DR-learner	100	2.006 ± 0.270	0.124 ± 0.009	ok	–	0.98 (1.119)	
BCF	100	2.666 ± 0.440	0.209 ± 0.027	ok	0.70 (3.397)	0.94 (0.935)	
causal forest	100	3.803 ± 0.624	0.542 ± 0.094	×	0.43 (3.188)	1.00 (4.196)	
TabPFN-DR	100	0.919 ± 0.151	0.129 ± 0.011	ok	–	0.99 (1.185)	
MP-OSPC	100	–	0.368 ± 0.057	×	–	0.97 (2.612)	
A-V5 (CausalPFN)	100	0.579 ± 0.070	0.143 ± 0.012	ok	–	0.99 (1.276); trim. 0.99	0.02
naive argmin (CausalPFN)	100	0.583 ± 0.070	–		–	–	
isotonic cal. (CausalPFN)	100	0.995 ± 0.093	–		–	–	
A-V5 (CausalFM)	100	2.105 ± 0.270	1.006 ± 0.064	×	–	0.68 (3.274); trim. 0.70	1.00
naive argmin (CausalFM)	100	1.974 ± 0.272	–		–	–	
isotonic cal. (CausalFM)	100	3.551 ± 0.544	–		–	–	

A-V5 vs. each method: paired difference of $\sqrt{\text{PEHE}}$ (A-V5 − method), 95% bootstrap interval, Holm-adjusted Wilcoxon p within benchmark and PFN

A-V5 of	vs.	k	diff	95% CI	Holm p	verdict
CausalPFN	CausalPFN	100	−0.0005	[−0.0014, +0.0000]	0.8325	no difference
CausalPFN	CausalPFN (temp.-cal.)	100	−0.0005	[−0.0014, +0.0000]	0.8325	no difference
CausalPFN	CausalFM	100	−5.2811	[−6.7047, −4.0868]	<0.0001	better
CausalPFN	Do-PFN	100	−5.4861	[−7.2595, −3.9873]	<0.0001	better
CausalPFN	DR-learner	100	−1.4272	[−1.8665, −1.0617]	<0.0001	better
CausalPFN	BCF	100	−2.0868	[−2.9042, −1.4135]	<0.0001	better
CausalPFN	causal forest	100	−3.2245	[−4.4248, −2.2119]	<0.0001	better
CausalPFN	TabPFN-DR	100	−0.3402	[−0.5240, −0.2013]	<0.0001	better
CausalPFN	naive argmin (CausalPFN)	100	−0.0035	[−0.0121, +0.0038]	0.8325	no difference
CausalPFN	isotonic cal. (CausalPFN)	100	−0.4163	[−0.5417, −0.3041]	<0.0001	better
CausalFM	CausalPFN	100	+1.5251	[+1.1572, +1.9660]	<0.0001	worse
CausalFM	CausalPFN (temp.-cal.)	100	+1.5251	[+1.1572, +1.9660]	<0.0001	worse
CausalFM	CausalFM	100	−3.7555	[−4.7678, −2.9092]	<0.0001	better
CausalFM	Do-PFN	100	−3.9604	[−5.3384, −2.8023]	<0.0001	better
CausalFM	DR-learner	100	+0.0984	[+0.0361, +0.1673]	0.0199	worse
CausalFM	BCF	100	−0.5611	[−0.9641, −0.2410]	0.0091	better
CausalFM	causal forest	100	−1.6989	[−2.4842, −1.0382]	<0.0001	better
CausalFM	TabPFN-DR	100	+1.1854	[+0.9327, +1.4780]	<0.0001	worse
CausalFM	naive argmin (CausalFM)	100	+0.1308	[+0.0651, +0.2049]	0.0012	worse
CausalFM	isotonic cal. (CausalFM)	100	−1.4468	[−2.1264, −0.8899]	<0.0001	better

Table 14. B2 ACIC-2016: every method on the same tasks. $\sqrt{\text{PEHE}}$ and |ATE err| as mean ± s.e.; sanity: × if the mean |ATE err| is worse than the naive difference in means (0.823); CATE 90% and ATE 95% coverage (mean width); deferred: share of tasks where A-V5 deferred in some region.

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
CausalPFN	770	1.424 ± 0.040	0.150 ± 0.006	ok	1.00 (12.5)	0.52 (0.226)	
CausalPFN (temp.-cal.)	770	1.424 ± 0.040	0.150 ± 0.006	ok	0.98 (5.629)	0.25 (0.103)	
CausalFM	770	4.227 ± 0.068	1.369 ± 0.029	×	0.85 (10.5)	–	
Do-PFN	770	5.092 ± 0.068	2.706 ± 0.040	×	–	–	
DR-learner	770	1.258 ± 0.041	0.117 ± 0.006	ok	–	0.80 (0.322)	
BCF	770	1.203 ± 0.043	0.100 ± 0.007	ok	0.75 (1.782)	0.79 (0.309)	
causal forest	770	1.619 ± 0.058	0.139 ± 0.010	ok	0.52 (1.295)	0.99 (1.986)	
TabPFN-DR	770	0.995 ± 0.045	0.101 ± 0.010	ok	–	0.86 (0.327)	
MP-OSPC	77	–	0.133 ± 0.015	ok	–	0.99 (1.016)	
A-V5 (CausalPFN)	770	1.220 ± 0.041	0.095 ± 0.005	ok	–	0.85 (0.345); trim. 0.95	0.60
naive argmin (CausalPFN)	770	1.174 ± 0.041	–		–	–	
isotonic cal. (CausalPFN)	770	1.549 ± 0.045	–		–	–	
A-V5 (CausalFM)	770	1.196 ± 0.041	0.224 ± 0.012	ok	–	0.64 (0.395); trim. 0.83	1.00
naive argmin (CausalFM)	770	1.191 ± 0.041	–		–	–	
isotonic cal. (CausalFM)	770	3.482 ± 0.063	–		–	–	

A-V5 vs. each method: paired difference of $\sqrt{\text{PEHE}}$ (A-V5 − method), 95% bootstrap interval, Holm-adjusted Wilcoxon p within benchmark and PFN

A-V5 of	vs.	k	diff	95% CI	Holm p	verdict
CausalPFN	CausalPFN	770	−0.2043	[−0.2350, −0.1742]	<0.0001	better
CausalPFN	CausalPFN (temp.-cal.)	770	−0.2043	[−0.2350, −0.1742]	<0.0001	better
CausalPFN	CausalFM	770	−3.0068	[−3.1238, −2.8912]	<0.0001	better
CausalPFN	Do-PFN	770	−3.8725	[−3.9939, −3.7521]	<0.0001	better
CausalPFN	DR-learner	770	−0.0387	[−0.0667, −0.0112]	0.0702	no difference
CausalPFN	BCF	770	+0.0169	[−0.0258, +0.0615]	0.7373	no difference
CausalPFN	causal forest	770	−0.3994	[−0.4954, −0.3221]	<0.0001	better
CausalPFN	TabPFN-DR	770	+0.2252	[+0.1810, +0.2643]	<0.0001	worse
CausalPFN	naive argmin (CausalPFN)	770	+0.0461	[+0.0244, +0.0683]	<0.0001	worse
CausalPFN	isotonic cal. (CausalPFN)	770	−0.3289	[−0.3773, −0.2827]	<0.0001	better
CausalFM	CausalPFN	770	−0.2277	[−0.2666, −0.1894]	<0.0001	better
CausalFM	CausalPFN (temp.-cal.)	770	−0.2277	[−0.2666, −0.1894]	<0.0001	better
CausalFM	CausalFM	770	−3.0303	[−3.1484, −2.9151]	<0.0001	better
CausalFM	Do-PFN	770	−3.8959	[−4.0177, −3.7771]	<0.0001	better
CausalFM	DR-learner	770	−0.0621	[−0.0848, −0.0417]	<0.0001	better
CausalFM	BCF	770	−0.0066	[−0.0486, +0.0360]	0.2589	no difference
CausalFM	causal forest	770	−0.4228	[−0.5175, −0.3467]	<0.0001	better
CausalFM	TabPFN-DR	770	+0.2018	[+0.1558, +0.2422]	<0.0001	worse
CausalFM	naive argmin (CausalFM)	770	+0.0054	[−0.0067, +0.0172]	0.0385	no difference
CausalFM	isotonic cal. (CausalFM)	770	−2.2859	[−2.3897, −2.1839]	<0.0001	better

Table 15. B2 subset (77): every method on the same tasks. $\sqrt{\text{PEHE}}$ and |ATE err| as mean ± s.e.; sanity: × if the mean |ATE err| is worse than the naive difference in means (0.747); CATE 90% and ATE 95% coverage (mean width); deferred: share of tasks where A-V5 deferred in some region.

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
CausalPFN	77	1.307 ± 0.089	0.143 ± 0.014	ok	1.00 (12.3)	0.51 (0.224)	
CausalPFN (temp.-cal.)	77	1.307 ± 0.089	0.143 ± 0.014	ok	0.98 (5.588)	0.21 (0.102)	
CausalFM	77	4.250 ± 0.228	1.312 ± 0.093	×	0.86 (10.4)	–	
Do-PFN	77	5.128 ± 0.223	2.672 ± 0.120	×	–	–	
DR-learner	77	1.171 ± 0.097	0.086 ± 0.011	ok	–	0.82 (0.321)	
BCF	77	1.123 ± 0.100	0.101 ± 0.015	ok	0.75 (1.764)	0.75 (0.302)	
causal forest	77	1.437 ± 0.103	0.107 ± 0.015	ok	0.53 (1.286)	1.00 (1.883)	
TabPFN-DR	77	0.900 ± 0.092	0.091 ± 0.020	ok	–	0.90 (0.334)	
MP-OSPC	77	–	0.133 ± 0.015	ok	–	0.99 (1.016)	
A-V5 (CausalPFN)	77	1.064 ± 0.077	0.076 ± 0.008	ok	–	0.87 (0.343); trim. 0.96	0.53
naive argmin (CausalPFN)	77	1.059 ± 0.075	–		–	–	
isotonic cal. (CausalPFN)	77	1.432 ± 0.092	–		–	–	
A-V5 (CausalFM)	77	1.097 ± 0.075	0.201 ± 0.028	ok	–	0.65 (0.405); trim. 0.86	1.00
naive argmin (CausalFM)	77	1.095 ± 0.080	–		–	–	
isotonic cal. (CausalFM)	77	3.561 ± 0.214	–		–	–	

A-V5 vs. each method: paired difference of $\sqrt{\text{PEHE}}$ (A-V5 − method), 95% bootstrap interval, Holm-adjusted Wilcoxon p within benchmark and PFN

A-V5 of	vs.	k	diff	95% CI	Holm p	verdict
CausalPFN	CausalPFN	77	−0.2428	[−0.3583, −0.1373]	<0.0001	better
CausalPFN	CausalPFN (temp.-cal.)	77	−0.2428	[−0.3583, −0.1373]	<0.0001	better
CausalPFN	CausalFM	77	−3.1862	[−3.6192, −2.7725]	<0.0001	better
CausalPFN	Do-PFN	77	−4.0639	[−4.4965, −3.6433]	<0.0001	better
CausalPFN	DR-learner	77	−0.1069	[−0.2662, +0.0133]	0.4027	no difference
CausalPFN	BCF	77	−0.0587	[−0.1989, +0.0759]	0.8992	no difference

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
CausalPFN	causal forest	77	−0.3725 [−0.5175, −0.2257]	<0.0001	better		
CausalPFN	TabPFN-DR	77	+0.1636 [+0.0463, +0.2722]	0.0004	worse		
CausalPFN	naive argmin (CausalPFN)	77	+0.0054 [−0.0608, +0.0638]	0.8756	no difference		
CausalPFN	isotonic cal. (CausalPFN)	77	−0.3684 [−0.4899, −0.2503]	<0.0001	better		
CausalFM	CausalPFN	77	−0.2100 [−0.3515, −0.0695]	0.0109	better		
CausalFM	CausalPFN (temp.-cal.)	77	−0.2100 [−0.3515, −0.0695]	0.0109	better		
CausalFM	CausalFM	77	−3.1534 [−3.5937, −2.7322]	<0.0001	better		
CausalFM	Do-PFN	77	−4.0311 [−4.4673, −3.6079]	<0.0001	better		
CausalFM	DR-learner	77	−0.0741 [−0.2002, +0.0017]	0.3328	no difference		
CausalFM	BCF	77	−0.0259 [−0.1690, +0.1158]	0.9801	no difference		
CausalFM	causal forest	77	−0.3397 [−0.4862, −0.1914]	<0.0001	better		
CausalFM	TabPFN-DR	77	+0.1964 [+0.0784, +0.3030]	<0.0001	worse		
CausalFM	naive argmin (CausalFM)	77	+0.0021 [−0.0369, +0.0316]	0.3328	no difference		
CausalFM	isotonic cal. (CausalFM)	77	−2.4641 [−2.8565, −2.0815]	<0.0001	better		

Table 16. B3 LaLonde-CPS: every method on the same tasks. $\sqrt{\text{PEHE}}$ (ITE-RMSE against noisy ITEs) and |ATE err| as mean ± s.e.; sanity: × if the mean |ATE err| is worse than the naive difference in means (2,889); CATE 90% and ATE 95% coverage (mean width); deferred: share of tasks where A-V5 deferred in some region.

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
CausalPFN	100	8,955 ± 20.9	1,002 ± 73.3	ok	–	0.11 (350)	
CausalPFN (temp.-cal.)	100	8,955 ± 20.9	1,002 ± 73.3	ok	–	0.01 (168)	
CausalFM	100	12,400 ± 15.8	6,625 ± 7.714	×	–	–	
Do-PFN	100	12,015 ± 32.0	6,182 ± 46.1	×	–	–	
DR-learner	100	9,173 ± 25.5	1,554 ± 90.1	ok	–	0.11 (766)	
BCF	100	12,765 ± 67.0	7,032 ± 97.8	×	–	0.00 (6,426)	
causal forest	100	1,342,574 ± 39,626	100,180 ± 7,773	×	–	1.00 (6,754,442)	
A-V5 (CausalPFN)	100	8,976 ± 21.6	934 ± 71.7	ok	–	0.21 (740); trim. 0.94	0.12
naive argmin (CausalPFN)	100	8,998 ± 22.8	–		–	–	
isotonic cal. (CausalPFN)	100	9,034 ± 22.5	–		–	–	
A-V5 (CausalFM)	100	9,167 ± 25.3	6,502 ± 22.2	×	–	0.00 (832); trim. 0.94	1.00
naive argmin (CausalFM)	100	9,173 ± 25.5	–		–	–	
isotonic cal. (CausalFM)	100	9,276 ± 19.9	–		–	–	

A-V5 vs. each method: paired difference of $\sqrt{\text{PEHE}}$ (A-V5 − method), 95% bootstrap interval, Holm-adjusted Wilcoxon p within benchmark and PFN

A-V5 of	vs.	k	diff	95% CI	Holm p	verdict
CausalPFN	CausalPFN	100	+21.1	[+5.0475, +40.6]	0.5637	no difference
CausalPFN	CausalPFN (temp.-cal.)	100	+21.1	[+5.0475, +40.6]	0.5637	no difference
CausalPFN	CausalFM	100	−3,425	[−3,468, −3,380]	<0.0001	better
CausalPFN	Do-PFN	100	−3,039	[−3,105, −2,976]	<0.0001	better
CausalPFN	DR-learner	100	−197	[−236, −159]	<0.0001	better
CausalPFN	BCF	100	−3,790	[−3,918, −3,659]	<0.0001	better
CausalPFN	causal forest	100	−1,333,599	[−1,412,082, −1,256,032]	<0.0001	better
CausalPFN	naive argmin (CausalPFN)	100	−22.3	[−40.8, −6.0829]	0.0036	better
CausalPFN	isotonic cal. (CausalPFN)	100	−58.5	[−82.8, −35.0]	<0.0001	better
CausalFM	CausalPFN	100	+212	[+173, +250]	<0.0001	worse
CausalFM	CausalPFN (temp.-cal.)	100	+212	[+173, +250]	<0.0001	worse
CausalFM	CausalFM	100	−3,234	[−3,287, −3,179]	<0.0001	better
CausalFM	Do-PFN	100	−2,848	[−2,920, −2,776]	<0.0001	better
CausalFM	DR-learner	100	−6.3932	[−9.7509, −3.2183]	0.0002	better
CausalFM	BCF	100	−3,599	[−3,746, −3,452]	<0.0001	better
CausalFM	causal forest	100	−1,333,408	[−1,411,925, −1,255,858]	<0.0001	better
CausalFM	naive argmin (CausalFM)	100	−6.3932	[−9.7509, −3.2183]	0.0002	better
CausalFM	isotonic cal. (CausalFM)	100	−110	[−141, −78.0]	<0.0001	better

Table 17. B4 LaLonde-PSID: every method on the same tasks. $\sqrt{\text{PEHE}}$ (ITE-RMSE against noisy ITEs) and |ATE err| as mean ± s.e.; sanity: × if the mean |ATE err| is worse than the naive difference in means (942); CATE 90% and ATE 95% coverage (mean width); deferred: share of tasks where A-V5 deferred in some region.

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
CausalPFN	100	14,383 ± 197	2,914 ± 219	×	–	0.08 (1,527)	
CausalPFN (temp.-cal.)	100	14,383 ± 197	2,914 ± 219	×	–	0.29 (820)	
CausalFM	100	22,436 ± 130	12,717 ± 31.3	×	–	–	
Do-PFN	100	20,907 ± 138	12,323 ± 86.0	×	–	–	
DR-learner	100	14,902 ± 279	1,069 ± 128	×	–	0.75 (3,206)	
BCF	100	22,553 ± 146	13,509 ± 115	×	–	0.00 (6,259)	
causal forest	100	48,314 ± 2,199	15,136 ± 724	×	–	1.00 (294,524)	
TabPFN-DR	100	13,757 ± 182	1,152 ± 126	×	–	0.68 (3,240)	
MP-OSPC	100	–	10,218 ± 252	×	–	0.00 (5,262)	
A-V5 (CausalPFN)	100	14,329 ± 196	2,400 ± 231	×	–	0.47 (4,032); trim. 1.00	0.09

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
naive argmin (CausalPFN)	100	14,315 ± 200	–		–	–	
isotonic cal. (CausalPFN)	100	14,101 ± 193	–		–	–	
A-V5 (CausalFM)	100	14,810 ± 250	11,540 ± 130	×	–	0.00 (5,064); trim. 0.93	0.97
naive argmin (CausalFM)	100	14,900 ± 255	–		–	–	
isotonic cal. (CausalFM)	100	18,545 ± 151	–		–	–	

A-V5 vs. each method: paired difference of $\sqrt{\text{PEHE}}$ (A-V5 − method), 95% bootstrap interval, Holm-adjusted Wilcoxon p within benchmark and PFN

A-V5 of	vs.	k	diff	95% CI	Holm p	verdict
CausalPFN	CausalPFN	100	−53.9	[−105, −13.7]	1.0000	no difference
CausalPFN	CausalPFN (temp.-cal.)	100	−53.9	[−105, −13.7]	1.0000	no difference
CausalPFN	CausalFM	100	−8,107	[−8,382, −7,804]	<0.0001	better
CausalPFN	Do-PFN	100	−6,579	[−6,834, −6,285]	<0.0001	better
CausalPFN	DR-learner	100	−574	[−919, −278]	<0.0001	better
CausalPFN	BCF	100	−8,225	[−8,473, −7,962]	<0.0001	better
CausalPFN	causal forest	100	−33,985	[−38,345, −29,860]	<0.0001	better
CausalPFN	TabPFN-DR	100	+572	[+405, +773]	<0.0001	worse
CausalPFN	naive argmin (CausalPFN)	100	+14.0	[−21.3, +55.8]	1.0000	no difference
CausalPFN	isotonic cal. (CausalPFN)	100	+227	[+114, +347]	0.0038	worse
CausalFM	CausalPFN	100	+427	[+148, +754]	0.0092	worse
CausalFM	CausalPFN (temp.-cal.)	100	+427	[+148, +754]	0.0092	worse
CausalFM	CausalFM	100	−7,626	[−8,031, −7,188]	<0.0001	better
CausalFM	Do-PFN	100	−6,097	[−6,487, −5,666]	<0.0001	better
CausalFM	DR-learner	100	−92.5	[−312, +182]	0.0006	no difference
CausalFM	BCF	100	−7,743	[−8,113, −7,357]	<0.0001	better
CausalFM	causal forest	100	−33,504	[−37,858, −29,412]	<0.0001	better
CausalFM	TabPFN-DR	100	+1,053	[+721, +1,446]	<0.0001	worse
CausalFM	naive argmin (CausalFM)	100	−89.8	[−140, −44.5]	0.0007	better
CausalFM	isotonic cal. (CausalFM)	100	−3,735	[−4,071, −3,359]	<0.0001	better

Table 18. H held-out: every method on the same tasks. $\sqrt{\text{PEHE}}$ and |ATE err| as mean ± s.e.; sanity: × if the mean |ATE err| is worse than the naive difference in means (4.586); CATE 90% and ATE 95% coverage (mean width); deferred: share of tasks where A-V5 deferred in some region.

method	k	$\sqrt{\text{PEHE}}$	|ATE err|	sanity	CATE cov. 90	ATE cov. 95	deferred
CausalPFN	240	2.130 ± 0.158	0.755 ± 0.079	ok	1.00 (34.6)	0.52 (1.543)	
CausalPFN (temp.-cal.)	240	2.130 ± 0.158	0.755 ± 0.079	ok	1.00 (15.9)	0.31 (0.645)	
CausalFM	240	3.839 ± 0.357	2.936 ± 0.340	ok	0.89 (15.3)	–	
Do-PFN	240	2.006 ± 0.108	0.850 ± 0.068	ok	–	–	
DR-learner	240	3.166 ± 0.287	0.968 ± 0.101	ok	–	0.53 (2.400)	
BCF	240	1.782 ± 0.192	0.813 ± 0.098	ok	0.56 (4.479)	0.57 (1.972)	
causal forest	240	2.035 ± 0.192	0.792 ± 0.099	ok	0.44 (3.532)	0.67 (6.196)	
TabPFN-DR	240	1.318 ± 0.089	0.825 ± 0.101	ok	–	0.64 (2.518)	
MP-OSPC	240	–	1.428 ± 0.330	ok	–	0.62 (5.360)	
A-V5 (CausalPFN)	240	1.901 ± 0.142	0.885 ± 0.108	ok	–	0.61 (2.854); trim. 0.60	0.30
naive argmin (CausalPFN)	240	1.684 ± 0.131	–		–	–	
isotonic cal. (CausalPFN)	240	2.645 ± 0.317	–		–	–	
A-V5 (CausalFM)	240	2.562 ± 0.262	0.955 ± 0.159	ok	–	0.61 (3.296); trim. 0.62	0.61
naive argmin (CausalFM)	240	1.878 ± 0.197	–		–	–	
isotonic cal. (CausalFM)	240	2.309 ± 0.140	–		–	–	

A-V5 vs. each method: paired difference of $\sqrt{\text{PEHE}}$ (A-V5 − method), 95% bootstrap interval, Holm-adjusted Wilcoxon p within benchmark and PFN

A-V5 of	vs.	k	diff	95% CI	Holm p	verdict
CausalPFN	CausalPFN	240	−0.2288	[−0.3608, −0.1399]	<0.0001	better
CausalPFN	CausalPFN (temp.-cal.)	240	−0.2288	[−0.3608, −0.1399]	<0.0001	better
CausalPFN	CausalFM	240	−1.9380	[−2.4548, −1.4958]	<0.0001	better
CausalPFN	Do-PFN	240	−0.1049	[−0.2710, +0.0853]	0.0038	no difference
CausalPFN	DR-learner	240	−1.2653	[−1.6302, −0.9325]	<0.0001	better
CausalPFN	BCF	240	+0.1189	[−0.1292, +0.3028]	<0.0001	no difference
CausalPFN	causal forest	240	−0.1338	[−0.3881, +0.1019]	0.2784	no difference
CausalPFN	TabPFN-DR	240	+0.5828	[+0.4298, +0.7740]	<0.0001	worse
CausalPFN	naive argmin (CausalPFN)	240	+0.2164	[+0.1384, +0.2971]	<0.0001	worse
CausalPFN	isotonic cal. (CausalPFN)	240	−0.7440	[−1.2490, −0.4149]	<0.0001	better
CausalFM	CausalPFN	240	+0.4325	[+0.1212, +0.7947]	0.8047	no difference
CausalFM	CausalPFN (temp.-cal.)	240	+0.4325	[+0.1212, +0.7947]	0.8047	no difference
CausalFM	CausalFM	240	−1.2766	[−1.6072, −0.9639]	<0.0001	better
CausalFM	Do-PFN	240	+0.5565	[+0.1790, +1.0284]	0.8047	no difference
CausalFM	DR-learner	240	−0.6039	[−0.9595, −0.2978]	<0.0001	better
CausalFM	BCF	240	+0.7803	[+0.5691, +1.0155]	<0.0001	worse
CausalFM	causal forest	240	+0.5275	[+0.2529, +0.8142]	<0.0001	worse
CausalFM	TabPFN-DR	240	+1.2442	[+0.8667, +1.7323]	<0.0001	worse
CausalFM	naive argmin (CausalFM)	240	+0.6841	[+0.4635, +0.9247]	<0.0001	worse
CausalFM	isotonic cal. (CausalFM)	240	+0.2531	[−0.1037, +0.6983]	<0.0001	no difference

Table 19. Held-out suite by mechanism and n: mean $\sqrt{\text{PEHE}}$ (DR: DR-learner, CF: causal forest); last column: coverage of A's 95% AIPW ATE interval for CausalPFN / CausalFM (descriptive; H6 has hidden confounding and is untestable).

mechanism	n	A-V5 (CausalPFN)	A-V5 (CausalFM)	CausalPFN	CausalFM	TabPFN-DR	BCF	DR	CF	ATE cov.
friedman-step	500	1.173	1.039	1.426	2.041	0.618	0.564	1.318	0.873	1.00 / 0.95
friedman-step	2000	0.454	0.636	1.050	1.910	0.305	0.315	0.540	0.358	1.00 / 0.95
heavy-mult	500	3.224	5.695	3.942	7.347	1.807	4.295	6.873	4.508	1.00 / 0.90
heavy-mult	2000	2.553	5.733	2.744	9.590	1.443	3.518	8.992	6.592	1.00 / 1.00
highcard-cat	500	1.065	0.987	1.142	1.112	0.530	0.844	1.034	0.975	0.80 / 0.85
highcard-cat	2000	0.425	0.430	0.985	1.034	0.209	0.407	0.526	0.692	0.70 / 0.80
iv-in-x	500	0.747	0.625	0.742	0.608	0.728	0.749	0.981	0.749	0.00 / 0.00
iv-in-x	2000	0.731	0.714	0.730	0.611	0.759	0.746	0.797	0.738	0.00 / 0.00
kang-schafer	500	6.534	8.884	6.564	10.279	4.947	5.078	9.895	4.361	0.95 / 1.00
kang-schafer	2000	4.207	4.176	4.533	9.570	2.861	3.291	5.113	2.868	0.90 / 0.90
mbias	500	0.894	0.967	0.894	0.997	0.847	0.805	1.092	0.896	0.00 / 0.00
mbias	2000	0.804	0.861	0.804	0.968	0.763	0.771	0.833	0.806	0.00 / 0.00

Table 20. Cost per task (wall seconds, 4 threads, fresh process): median [IQR], peak memory (MB, median), and A-V5's overhead relative to the plain PFN (median per-task ratio; A-V5's cost counts the $K + 1$ PFN passes and the reference bundle whether or not the cache was hit). The B2 TabPFN-DR and MP-OSPC jobs ran at several load levels, before and after a pause of the run (Appendix A).

set	method	k	wall	memory	overhead
B1 IHDP	CausalPFN	100	9.080 [7.012, 10.8]	1,737	
B1 IHDP	CausalPFN (temp.-cal.)	100	49.1 [46.6, 52.5]	1,737	
B1 IHDP	CausalFM	100	10.2 [8.320, 11.3]	1,737	
B1 IHDP	Do-PFN	100	6.098 [5.490, 7.795]	1,737	
B1 IHDP	DR-learner	100	27.5 [26.6, 30.0]	1,737	
B1 IHDP	BCF	100	23.9 [23.0, 27.8]	1,737	
B1 IHDP	causal forest	100	18.0 [17.6, 19.8]	1,737	
B1 IHDP	TabPFN-DR	100	157 [142, 175]	1,737	
B1 IHDP	MP-OSPC	100	301 [289, 319]	2,565	
B1 IHDP	A-V5 (CausalPFN)	100	239 [230, 256]		26.8×
B1 IHDP	A-V5 (CausalFM)	100	261 [253, 277]		27.2×
B2 ACIC-2016	CausalPFN	770	89.3 [84.2, 93.5]	1,884	
B2 ACIC-2016	CausalPFN (temp.-cal.)	770	401 [389, 414]	2,212	
B2 ACIC-2016	CausalFM	770	185 [178, 192]	3,492	
B2 ACIC-2016	Do-PFN	770	83.0 [78.9, 86.4]	1,884	
B2 ACIC-2016	DR-learner	770	93.6 [91.3, 99.4]	1,884	
B2 ACIC-2016	BCF	770	84.0 [81.9, 87.6]	1,884	
B2 ACIC-2016	causal forest	770	42.7 [41.1, 44.9]	1,886	
B2 ACIC-2016	TabPFN-DR	770	4,463 [2,938, 4,834]	2,782	
B2 ACIC-2016	MP-OSPC	77	2,882 [2,686, 3,050]	5,655	
B2 ACIC-2016	A-V5 (CausalPFN)	770	1,246 [1,216, 1,287]		14.2×
B2 ACIC-2016	A-V5 (CausalFM)	770	1,631 [1,597, 1,682]		8.8×
B3 LaLonde-CPS	CausalPFN	100	106 [97.6, 159]	1,737	
B3 LaLonde-CPS	CausalPFN (temp.-cal.)	100	1,232 [1,214, 1,266]	4,804	
B3 LaLonde-CPS	CausalFM	100	260 [245, 268]	2,155	
B3 LaLonde-CPS	Do-PFN	100	818 [778, 872]	10,279	
B3 LaLonde-CPS	DR-learner	100	84.1 [82.0, 88.5]	1,737	
B3 LaLonde-CPS	BCF	100	176 [171, 183]	1,737	
B3 LaLonde-CPS	causal forest	100	52.3 [51.1, 53.3]	1,737	
B3 LaLonde-CPS	A-V5 (CausalPFN)	100	1,720 [1,658, 1,774]		15.0×
B3 LaLonde-CPS	A-V5 (CausalFM)	100	2,535 [2,493, 2,580]		9.9×
B4 LaLonde-PSID	CausalPFN	100	46.3 [41.3, 49.0]	1,737	
B4 LaLonde-PSID	CausalPFN (temp.-cal.)	100	216 [210, 222]	1,737	
B4 LaLonde-PSID	CausalFM	100	26.8 [23.6, 28.2]	1,737	
B4 LaLonde-PSID	Do-PFN	100	41.6 [39.0, 43.2]	1,737	
B4 LaLonde-PSID	DR-learner	100	57.7 [55.0, 63.0]	1,737	
B4 LaLonde-PSID	BCF	100	55.3 [53.7, 61.2]	1,737	
B4 LaLonde-PSID	causal forest	100	27.9 [26.6, 31.0]	1,737	
B4 LaLonde-PSID	TabPFN-DR	100	389 [379, 395]	1,791	
B4 LaLonde-PSID	MP-OSPC	100	422 [413, 429]	1,737	
B4 LaLonde-PSID	A-V5 (CausalPFN)	100	650 [626, 678]		14.1×
B4 LaLonde-PSID	A-V5 (CausalFM)	100	671 [653, 698]		26.0×
H held-out	CausalPFN	240	16.7 [6.702, 24.3]	1,737	
H held-out	CausalPFN (temp.-cal.)	240	97.8 [43.5, 173]	1,737	
H held-out	CausalFM	240	11.1 [9.234, 17.1]	1,737	
H held-out	Do-PFN	240	16.3 [10.7, 26.9]	1,737	
H held-out	DR-learner	240	41.8 [24.7, 59.1]	1,737	
H held-out	BCF	240	38.3 [21.0, 50.8]	1,737	
H held-out	causal forest	240	22.3 [17.7, 27.8]	1,737	
H held-out	TabPFN-DR	240	187 [95.0, 323]	1,737	
H held-out	MP-OSPC	240	263 [146, 354]	1,737	
H held-out	A-V5 (CausalPFN)	240	357 [194, 505]		23.3×
H held-out	A-V5 (CausalFM)	240	374 [207, 535]		27.0×

Table 22. Exploratory follow-up (written after the confirmatory verdicts; no claim uses it). Top: A-V5's deferring tasks, the harmful ones, the size of the harm (relative change of $\sqrt{\text{PEHE}}$ against always-PFN, median / 90th percentile), the naive argmin's switches and harmful switches on the same tasks, and failure events with a ratio $\sqrt{\text{PEHE}}(f)/\sqrt{\text{PEHE}}(\text{DR})$ below 1.25. Middle: deferring and harmful tasks by region and reference. Bottom: H by mechanism (H6: instruments in X and hidden confounding).

	H, CausalPFN	H, CausalFM	B2, CausalPFN	B2, CausalFM
deferring / harmful	72 / 4	147 / 19	463 / 97	770 / 11
harm size	+7% / +9%	+22% / +44%	+11% / +63%	+15% / +64%
naive argmin: switch / harmful	173 / 33	222 / 39	619 / 134	770 / 14
failure events (ratio < 1.25)	62 (15)	132 (33)	440 (109)	746 (8)
G_{in}: BCF	60 / 3	117 / 15	78 / 5	121 / 0
G_{in}: DR	4 / 0	11 / 2	132 / 12	194 / 1
G_{out}: BCF	3 / 1	1 / 0	9 / 0	–
G_{out}: DR	–	2 / 1	102 / 65	–
G_{in}: BCF + G_{out}: BCF	5 / 0	10 / 0	4 / 0	30 / 0
G_{in}: BCF + G_{out}: DR	–	2 / 0	57 / 8	172 / 7
G_{in}: DR + G_{out}: BCF	–	4 / 1	1 / 0	32 / 2
G_{in}: DR + G_{out}: DR	–	–	80 / 7	221 / 1
H1 kang-schafer, $n = 500$	2 / 0 of 20	6 / 0 of 20		
H1 kang-schafer, $n = 2000$	6 / 0 of 20	20 / 0 of 20		
H2 friedman-step, $n = 500$	9 / 0 of 20	19 / 0 of 20		
H2 friedman-step, $n = 2000$	20 / 0 of 20	20 / 0 of 20		
H3 heavy-mult, $n = 500$	3 / 1 of 20	4 / 0 of 20		
H3 heavy-mult, $n = 2000$	1 / 0 of 20	13 / 0 of 20		
H4 highcard-cat, $n = 500$	6 / 0 of 20	12 / 1 of 20		
H4 highcard-cat, $n = 2000$	20 / 0 of 20	20 / 0 of 20		
H5 mbias, $n = 500$	1 / 0 of 20	4 / 1 of 20		
H5 mbias, $n = 2000$	0 / 0 of 20	11 / 0 of 20		
H6 iv-in-x, $n = 500$	2 / 2 of 20	3 / 3 of 20		
H6 iv-in-x, $n = 2000$	2 / 1 of 20	15 / 14 of 20		

Table 21. Seed robustness (§4 of the protocol): mean $\sqrt{\text{PEHE}}$ with method seeds 0, 1, 2, and for A-V5 the share of tasks whose deferral decision agrees with seed 0.

set	method	k	seed 0	seed 1	seed 2	agreement (s1 / s2)
B1	A-V5 (CausalPFN)	100	0.579	0.579	0.579	0.99 / 0.99
B1	A-V5 (CausalFM)	100	2.105	2.049	2.035	1.00 / 1.00
B1	TabPFN-DR	100	0.919	0.941	0.945	
B1	BCF	100	2.666	2.651	2.670	
H n=500	A-V5 (CausalPFN)	120	2.273	2.292	2.275	0.83 / 0.83
H n=500	A-V5 (CausalFM)	120	3.033	3.112	2.934	0.83 / 0.84
H n=500	TabPFN-DR	120	1.579	1.612	1.654	
H n=500	BCF	120	2.056	1.930	1.827	

H. Post-Verdict Analyses Requested in Review

A simulated review of the submitted version asked for further analyses. All of them follow a plan committed before any of them was computed (pushed 2026-10-13 05:19 UTC). Its one addendum, made after R-A and before any R-C job finished, adds the R-loss to the repeated cross-fitting. The analyses are exploratory: the final benchmarks are reused for analysis only, no pre-registered verdict changes, and every variant is reported. The development pilots are analysed with the same code, so each table also shows whether the pilots alone would have pointed the same way. New jobs run through the evaluation harness from a frozen code version, on CPU. A second simulated review led to R2-A, R2-B and R2-E below, whose plan was pushed on 2026-10-14 at 20:36 UTC before any of them was computed, and to the confirmatory run of Appendix I.

R-A: risk scores and G_{out} (Table 23). The *R-loss* score is $\delta_i = (\tilde{Y}_i - \tilde{T}_i g_i)^2 - (\tilde{Y}_i - \tilde{T}_i r_i)^2$ with $\tilde{Y} = Y - \hat{m}$, $\tilde{T} = T - \hat{e}$ and $\hat{m} = \hat{e}\hat{m}_1 + (1-\hat{e})\hat{m}_0$ from the scoring nuisances (Nie & Wager, 2021). With correct nuisances its mean is the overlap-weighted excess $\mathbb{E}[e(1-e)\{(g-\tau)^2 - (r-\tau)^2\}]$, and it never divides by $\hat{e}$. The *overlap-weighted DR* score multiplies Equation (3) by $\hat{e}(1-\hat{e})$. *DR-learner by R-loss* scores the DR-learner by the R-loss and BCF by Equation (3), so no reference is judged by a criterion it was fitted to, and it chooses among rejected references by z. G_{in} *only* tests and

defers in G_{in} and keeps f in G_{out}, reporting "insufficient information" there. The naive rows switch to the reference with the lowest estimated risk, without a test.

For CausalPFN, the R-loss and the G_{in}-only test bring the pooled harm under the 10% bound (9.8% and 7.6%, against 18.9%), and the overlap-weighted DR score does not (18.3%). Because φ uses the same clipped $\hat{e}$, the weight cancels its inverse-propensity factors, $\hat{e}(1-\hat{e})\varphi = \hat{e}(1-\hat{e})(\hat{m}_1 - \hat{m}_0) + T(1-\hat{e})(Y - \hat{m}_1) - (1-T)\hat{e}(Y - \hat{m}_0)$, so this score has no heavy tails to remove; R2-A shows that it keeps the bias of the DR score. For CausalFM, both give up part of the repair on B2 and nearly all of it on LaLonde, where almost every unit has $\hat{e} < 0.1$ (on average 2% and 17% of the units lie in G_{in}), so that $\hat{e}(1-\hat{e})$ is close to 0. The naive argmin is the most accurate on H and B2 under every score, but where the PFN is best its error reaches 1.226× the PFN's (B1, R-loss), and it fails P-2's in-prior bound on the development data.

R-B: empirical size and the source of harm (Table 24). With the true CATE on the training units, we know which nulls hold. For CausalPFN the size holds on H (3.3% of the tasks with a true null have one rejected) and on the development platform (1.5%). It fails on B2 (28.0%), mostly through the G_{out} test against the DR-learner (109/249 true nulls rejected). This is where Assumption A1 fails: in 537 of 770 B2 tasks some true propensities lie outside $[0.01, 0.99]$, on average 11% of the units and 70% of those in G_{out}. Among CausalPFN's B2 deferrals of a single region, the harm rate is 65/79 when the oracle excess is ≤ 0 and 5/152 when it exceeds three standard errors: the harmful deferrals are test errors, not near-ties. For CausalFM the size fails mainly where the assumptions are violated. On H, 18/40 of the H6 tasks with a true null have one rejected (hidden confounding: φ is centred on a confounded contrast). On the platform, 9/13 of the hard-overlap tasks do (Assumption A1 fails). On B2 few nulls are true, because CausalFM is far worse than both references. The bound of Proposition 5.2(i) fails accordingly. For CausalPFN on B2, 121 deferrals follow a false rejection, 80 of them harmful, against α times the number of tasks with a true null, ≈ 22.

R2-A: bias or noise? (Table 31). With μ_0, μ_1 and e known, each paired statistic minus its oracle excess splits exactly into a bias B, the mean over units of the score's conditional mean given X and the out-of-fold nuisances minus the target, and a mean-zero noise N. For the DR score, $\mathbb{E}[\varphi \mid X] - \tau = (\hat{m}_1 - \mu_1)(1 - e/\hat{e}) - (\hat{m}_0 - \mu_0)\{1 - (1-e)/(1-\hat{e})\}$. For CausalPFN on B2, the G_{out} test against the DR-learner rejects 109 of its 249 true nulls; with B removed it would reject 2, with N removed 98. The median B is +2.95 standard errors, and +2.66 remain when the true propensity, clipped at 0.01, replaces $\hat{e}$. Most of the bias therefore comes from the truncation itself: where $e \ll 0.01 \leq \hat{e}$, the factor $1 - e/\hat{e}$ is close to 1, and the error of the outcome model in the nearly empty treated arm enters at first order instead of as a product of two errors. The bias favours deferral in 82% of these tests, against 45% for BCF: the DR-learner is fitted to pseudo-outcomes of the same form, so its error leans the same way and the score prefers it (Curth & van der Schaar, 2023). The overlap weight shrinks the noise but not the bias, which sits in $\mathbb{E}[\varphi \mid X]$ (92 rejections, 3 without B). The R-loss is not truncated, and its bias, $(g-r)\{(e-\hat{e})^2(g+r) - 2(e-\hat{e})(m-\hat{m})\}$ with $m = \mathbb{E}[Y \mid X]$, involves the error of the marginal regression, which the dominant arm pins down. Its median B is +0.14 standard errors (31 rejections, 4 without B). On H and the platform, B is small and the size holds. The failure thus calls for a score that is robust to bias where overlap fails, not for a test that is robust to heavy tails.

R2-B: a margin for the true null (Table 32). R-B calls a test's null true when its oracle excess is ≤ 0. This in-sample excess of the out-of-fold predictions is the statistic's target given X, but near-ties could be misclassified. With a margin, a null counts as true only when the oracle excess is at most $-k$ standard errors. For the registered test on B2, the G_{out} test against the DR-learner then rejects 82/148 clear nulls at $k = 1$ and 67/101 at $k = 2$, and 66/97 and 54/97 of the harmful deferrals still follow such a rejection. The clearer the null, the more often it is rejected, as a bias predicts, so R-B's conclusion stands. The false rejections of the R-loss are near-ties of its overlap-weighted estimand: at $k = 0.5$ it rejects 8/405 nulls (5/36 for the G_{out} test against the DR-learner), against 46/893 at $k = 0$.

R-C: repeated cross-fitting (Table 29). We redo the cross-fitting with new fold assignments: 20 splits for H at $n = 500$, and 10 for H at $n = 2000$ and for the B2 subset. Each test is aggregated over splits by the median rule of Chernozhukov et al. (2018), $\hat{\Delta} = \text{med}_s\, \hat{\Delta}_s$ and $\widehat{\text{se}}^2 = \text{med}_s\{\widehat{\text{se}}_s^2 + (\hat{\Delta}_s - \hat{\Delta})^2\}$, before Holm and the same choice rule. The deployed predictions are the same in every split, so decisions differ only through the folds. On H at $n = 500$, two independent single splits agree on whether to defer in 83% of the tasks. Two independent sets of 5 splits agree in 92%, and two sets of 10 in 92% (CausalFM: 83%, 90%, 96%). The aggregated test is also more conservative. With 10 splits, CausalPFN on H defers in 23% of the tasks instead of 30%. Its harmful share falls from 5.6% to 1.8%, and its mean $\sqrt{\text{PEHE}}$ rises from 1.901 to 1.943, still below always-CausalPFN (2.130). CausalFM's harmful share on H does not move (12.9% and 13.4%). Of its

18 harmful deferrals, 16 are in H6, where hidden confounding shifts φ, so more splits cannot help. The R-loss with 10 splits gives CausalPFN a harmful share of 1.6% on H (mean $\sqrt{\text{PEHE}}$ 1.983). Aggregating the per-split Holm-adjusted p-values instead (Meinshausen et al., 2009) gives 0.0% (1.940). On the B2 subset (A-1 is evaluated on B2), 10 splits take CausalPFN's harmful share from 14.6% to 17.5% (R-loss: 11.4%) and the AUROC of Z_{DR} from 0.836 to 0.826 (one split on all of B2: 0.783). Here the decision is already stable: two single splits agree on whether to defer in 92% of the tasks, and two sets of 5 in 99%. The harmful deferrals on B2 thus repeat across splits, as expected if they come from the size failure of the G_{out} test under low overlap (R-B) rather than from fold noise. With 10 splits the R-loss lowers the harmful share and lifts the AUROC to 0.899, but it defers in fewer tasks (45% against 52%) and has a higher mean $\sqrt{\text{PEHE}}$ (1.104 against 1.090). CausalFM defers on every subset task in every variant, and only 2 of them are harmful. On this evidence, repeated cross-fitting would not change A-1 or A-2. On B2 it moves neither the AUROC nor the harmful share toward the criterion, and in the registered run B2 contributes 97 of CausalPFN's 101 pooled harmful deferrals.

R-D: TabPFN-DR as a reference (Table 30). We cross-fit TabPFN-DR on the folds of the registered run and either add it to the references ($\mathcal{R} = \{\text{DR, BCF, TabPFN-DR}\}$, six Holm tests) or use it alone. The deployed predictions are those of the final run, and the A-V5 column reproduces the registered values. Adding TabPFN-DR improves A-V5 wherever TabPFN-DR is better than the PFN, but closes the gap to always using TabPFN-DR only on the B2 subset. There, CausalPFN reaches 0.924, against 1.064 without TabPFN-DR and 0.900 for TabPFN-DR alone (difference +0.023 [-0.062, +0.099]), choosing TabPFN-DR in 64% of the tasks. Its harmful share, 12%, stays above 10%. On H, CausalPFN reaches 1.800, against 1.901 and 1.318 (difference +0.482 [+0.336, +0.667]), with TabPFN-DR chosen in 26% of the tasks. For CausalFM on B1 it is 1.097, against 2.105 and 0.919. The test defers only on significant evidence, so it forgoes the gain wherever TabPFN-DR is better by a margin it cannot resolve. Where the PFN is better, the wrapper keeps it: on B1, CausalPFN (0.579, against 0.919 for TabPFN-DR) defers in 2% of the tasks and keeps its error (0.579). On B4, where most units lie in G_{out}, CausalPFN defers to TabPFN-DR in 51% of the tasks, and 8/51 of its deferrals are harmful.

R-E: how CausalFM and Do-PFN are used (Tables 27 and 28).

- *Code path.* Our CausalFM call is the toolkit's `PerFeatureTransformerCATE.estimate_cate` with the released standard-CATE checkpoint and the toolkit's head configuration. Our Do-PFN CATE is the package's `predict_cate`: the difference of predictive means with the treatment in the first column. The one deliberate difference is CausalFM's outcome scaler, which we fit on observed outcomes rather than on $y_0 \cup y_1$.
- *Published numbers.* On the ACIC-2016 subset that CausalFM's authors use, in their normalized units, our errors match theirs for both models. Their table also reports a DR-learner at 1.485, against our 0.156, which explains why the comparison there favours the PFNs.
- *Shrinkage.* On B1 both models estimate a heavily shrunken CATE (median slope 0.18 and 0.09, against 1.00 for CausalPFN) and a shrunken ATE (median ratio to the truth 0.45 and 0.80). This is why their |ATE err| (2.352, 1.910) is worse than the difference in means (0.286). On the platform their error grows with the effect size and heterogeneity while CausalPFN's and BCF's do not, and Do-PFN shrinks even at $d = 5$, inside its pretraining range (slope 0.23). We read this as a property of the models, not of how we call them; Do-PFN's authors note that their prior concentrates on small effects.
- *Preprocessing.* Table 28 (bottom) gives CausalFM under the toolkit's own scaling, run as a diagnostic only because it uses counterfactual outcomes. On B1 it lowers CausalFM's $\sqrt{\text{PEHE}}$ from 5.860 to 4.582 and removes the ATE shrinkage (median ratio 0.45 $\rightarrow$ 1.01). The toolkit's X scaler alone does not do this (5.855; ratio 0.45): the outcome scaler fitted on $y_0 \cup y_1$ centres y on the mean of both potential outcomes, which carries the ATE into the input. This leak is not available to a user. Even with it, CausalFM stays far behind the references on B1 (4.582 against 2.006 for the DR-learner and 2.666 for BCF), so its repair by certify-or-defer does not rest on our preprocessing.
- *Authors.* We have not yet asked the authors to confirm our use.

R-F: detection beyond the binary AUROC (Table 25). Z_{DR} is correlated with the continuous excess (Spearman 0.59–0.72), so the AUROC threshold hides a graded signal. Detection aligned with the decision rule, using failure relative to the better reference and the larger of the two statistics, reaches 0.833 for CausalPFN on B2, above A-1's 0.80, but 0.736 for CausalFM. On B2 the harm share of CausalPFN's switches stays above 10% at every threshold in the table, while on H a threshold of $t = 3$ removes it.

Table 23. Post-verdict analysis R-A (exploratory): risk scores and G_{out} handling. Mean $\sqrt{\text{PEHE}}$ relative to always-PFN on each final benchmark; harmful / deferring tasks pooled over H and B2 (the A-2 statistic; bound 10%); B2 AUROC of the full-sample statistic of that score for the failure event; and, on the development pilots (P01 + P02 pooled), the largest in-prior ratio, the share of the oracle gain recovered over the failure cells and the harm rate (the P-2 quantities; pass: ≤ 1.02, ≥ 0.5, $\leq 10\%$). Naive rows switch whenever an estimated risk is lower, without a test.

	variant	B1	B2	B3	B4	H	harm H+B2	AUROC B2	dev. ratio	dev. gain	dev. harm
CausalPFN	A-V5 (registered)	0.999	0.857	1.002	0.996	0.893	101/535 (18.9%)	0.783	1.012	0.56	8.6%
	R-loss score	0.999	0.850	1.000	1.000	0.887	47/479 (9.8%)	0.839	1.000	0.60	4.3%
	overlap-weighted DR	0.999	0.855	1.000	0.999	0.896	98/536 (18.3%)	0.826	1.012	0.58	10.0%
	DR-learner by R-loss	0.999	0.851	1.000	1.000	0.894	56/485 (11.5%)		1.001	0.54	5.8%
	test G_{in} only	0.999	0.850	1.000	1.000	0.884	35/462 (7.6%)		1.010	0.57	5.5%
	naive argmin, DR	1.005	0.824	1.005	0.995	0.791	167/792 (21.1%)		1.092	0.68	40.4%
	naive argmin, R-loss	1.226	0.785	1.074	1.033	0.760	136/794 (17.1%)		1.064	0.75	32.2%
	naive argmin, OW-DR	0.998	0.805	1.001	1.000	0.763	149/793 (18.8%)		1.072	0.71	36.7%
	naive region-wise argmin	1.153	0.825	1.006	0.995	0.801	225/897 (25.1%)		1.104	0.65	47.6%
CausalFM	A-V5 (registered)	0.359	0.283	0.739	0.660	0.667	30/917 (3.3%)	0.788	0.726	0.72	7.1%
	R-loss score	0.341	0.332	0.997	0.999	0.622	29/928 (3.1%)	0.837	0.665	0.78	7.3%
	overlap-weighted DR	0.349	0.283	0.739	0.679	0.670	31/919 (3.4%)	0.838	0.715	0.76	7.4%
	DR-learner by R-loss	0.453	0.294	0.987	0.991	0.677	18/917 (2.0%)		0.689	0.74	3.6%
	test G_{in} only	0.668	0.404	1.000	1.000	0.644	23/924 (2.5%)		0.703	0.72	6.6%
	naive argmin, DR	0.337	0.282	0.740	0.664	0.489	53/992 (5.3%)		0.656	0.83	19.1%
	naive argmin, R-loss	0.335	0.270	0.998	0.980	0.478	45/1001 (4.5%)		0.624	0.89	17.5%
	naive argmin, OW-DR	0.335	0.276	0.740	0.701	0.484	49/995 (4.9%)		0.624	0.86	18.1%
	naive region-wise argmin	0.341	0.282	0.739	0.651	0.503	56/1003 (5.6%)		0.644	0.80	21.2%

R-G: the two significance criteria (Table 26). The protocol makes the conjunction of the two criteria the decision rule and names no single primary test, so every comparison in the table is reported as no difference. The disagreements have one cause. The bootstrap tests the mean, while the Wilcoxon test, which drops zero differences, tests the location of the nonzero ones. On B3 and B4, A-V5 defers on few tasks; on the others it differs from the separate plain CausalPFN run only by numerical noise (below 10^{-4} in dollars), which dominates the Wilcoxon test, while the few deferrals decide the mean. On H against BCF, A-V5 is worse on most tasks but much better on a few, so the median and the mean point in opposite directions.

R2-E: software or model? Some diagnostic findings concern the released software and would change if the maintainers fixed it. CausalPFN's silent SVD for $d > 99$ and its context cap could be made visible or lifted; lifting the cap removes the loss at $n = 8000$ (Section 4.4). Do-PFN's feature limit could be enforced, which would block it for $d > 6$ rather than make it accurate there, and it shrinks the effect even at $d = 5$ (slope 0.23). CausalPFN's IHDP loader could use the per-realization truth. The other findings are properties of the models or of identification: the accuracy ranking, intervals that reflect outcome noise, signals blind to violated overlap, and a confounding bias that no estimator using (X, T, Y) removes. CausalPFN's ATE undercoverage comes from how the output is formed, from independent per-unit draws, which a maintainer could change; we have not tested such a fix. On the 46 platform cells that neither CausalPFN limit touches (all but $d = 100, 150$ and $n = 4000, 8000$), TabPFN-DR is still significantly better than the best classical method in 33 cells and worse in 2. It is better than CausalPFN in 37 and worse in 0, and it is the most accurate method in 35 (all 50 cells: 37, 2, 41, 0 and 39).

What the assumptions cover. Assumption A1 fails in the hard-overlap cells, and nearly so on LaLonde-CPS (1% treated); Assumption A2 fails for H3 (t_3 noise); identification fails in the hidden-confounding cells, H5 and H6. ACIC-2016 violates Assumption A1 for about a tenth of its units, most of them in G_{out}, which is where the size failed, through the bias of the DR score (R2-A). Weak overlap without violation ($\kappa = 3$) satisfies it with a small c, but its pseudo-outcomes are heavy-tailed enough that the normal approximation is poor at our sample sizes.

Table 24. Post-verdict analysis R-B (exploratory): empirical size and the source of harm. With the true CATE on the training units, a test's null $\bar{\Delta}_{g,r} \leq 0$ is known to hold when the oracle in-sample excess of the out-of-fold predictions is ≤ 0 (for the R-loss: the overlap-weighted excess with the true propensity). Columns: true nulls rejected after Holm, per test and pooled; tasks with a true null and at least one rejected (nominal $\alpha = 5\%$); harmful deferrals that follow a false rejection, and those with only true rejections whose order flips from the out-of-fold to the full-data fits on the test units.

set	G_{in}:DR	G_{in}:BCF	G_{out}:DR	G_{out}:BCF	all tests	tasks	harm: false rej.	harm: flip
B2, CausalPFN	12/257	3/242	109/249	14/179	138/927 (14.9%)	123/439 (28.0%)	80	17
R-loss	5/246	3/234	31/227	7/186	46/893 (5.2%)	42/442 (9.5%)	16	25
G_{in} only	16/257	4/242	–	–	20/499 (4.0%)	20/342 (5.8%)	14	13
H, CausalPFN	1/175	4/54	1/126	1/45	7/400 (1.8%)	6/184 (3.3%)	4	0
dev., CausalPFN	1/399	0/225	5/345	2/176	8/1145 (0.7%)	6/412 (1.5%)	5	1
B2, CausalFM	3/7	–	28/38	0/3	31/48 (64.6%)	31/45 (68.9%)	9	2
R-loss	0/1	–	19/52	4/13	23/66 (34.8%)	21/58 (36.2%)	5	5
H, CausalFM	3/98	17/40	2/94	1/43	23/275 (8.4%)	21/127 (16.5%)	19	0
dev., CausalFM	5/210	9/87	24/217	0/85	38/599 (6.3%)	35/263 (13.3%)	19	1

Table 25. Post-verdict analysis R-F (exploratory): detection beyond the binary AUROC. Spearman correlation of Z_{DR} with $\log(\sqrt{\text{PEHE}}_f / \sqrt{\text{PEHE}}_{\text{DR}})$, and of the largest regional z of A-V5 with $\log(\sqrt{\text{PEHE}}_f / \min_r \sqrt{\text{PEHE}}_r)$; AUROC of Z_{BCF} for failure relative to BCF, and of $\max(Z_{\text{DR}}, Z_{\text{BCF}})$ for failure relative to the better reference (events in brackets); decision curve: switching the whole output to the reference with the larger z when $\max z > t$, mean $\sqrt{\text{PEHE}}$ relative to always-PFN / share of switches that harm, at $t = 0, 1.645, 3$.

set	$\rho(Z_{\text{DR}})$	$\rho(\max z)$	AUROC vs BCF	AUROC vs best	$t = 0$	$t = 1.645$	$t = 3$
H, CausalPFN	0.66	0.65	0.777 (143)	0.786 (143)	0.79 / 18%	0.86 / 10%	0.95 / 0%
H, CausalFM	0.72	0.61	0.618 (198)	0.621 (198)	0.50 / 17%	0.57 / 15%	0.73 / 11%
B2, CausalPFN	0.59	0.47	0.817 (470)	0.833 (574)	0.83 / 21%	0.84 / 16%	0.86 / 13%
B2, CausalFM	0.67	0.52	0.682 (766)	0.736 (767)	0.29 / 1%	0.29 / 1%	0.29 / 1%

Table 26. Post-verdict analysis R-G: every A-V5 comparison in which exactly one of the two pre-specified criteria holds (Holm-adjusted Wilcoxon $p < 0.05$; bootstrap interval of the mean difference excluding 0). The rule of Section 3 requires both, so all of these are reported as no difference. Mean and median paired difference of $\sqrt{\text{PEHE}}$ (A-V5 minus the comparator) and the share of tasks on which A-V5 is better / worse.

bench	PFN	vs	mean diff [95% CI]	Holm p	median diff	better / worse
B2	CausalPFN	DR-learner	-0.0387 [-0.0667, -0.0112]	0.0702	+0	43% / 37%
B2	CausalFM	naive argmin	+0.00544 [-0.0067, +0.0172]	0.0385	+0	16% / 22%
B3	CausalPFN	CausalPFN	+21.1 [+5.05, +40.6]	0.564	-3.64e-07	58% / 42%
B4	CausalPFN	CausalPFN	-53.9 [-105, -13.7]	1	+0	49% / 47%
B4	CausalFM	DR-learner	-92.5 [-312, +182]	0.000605	-46.9	67% / 33%
H	CausalPFN	Do-PFN	-0.105 [-0.271, +0.0853]	0.00381	-0.0435	53% / 47%
H	CausalPFN	BCF	+0.119 [-0.129, +0.303]	1.32e-12	+0.0619	25% / 71%
H	CausalFM	CausalPFN	+0.433 [+0.121, +0.795]	0.805	-0.0282	56% / 44%
H	CausalFM	Do-PFN	+0.556 [+0.179, +1.03]	0.805	+0.0138	49% / 51%
H	CausalFM	isotonic cal.	+0.253 [-0.104, +0.698]	4.45e-06	-0.148	71% / 29%

Table 27. Usage check R-E. Left: ACIC-2016 (the causallib 10 datasets), $\sqrt{\text{PEHE}}$ divided by $\text{sd}(y_0 \cup y_1)$ as in CausalFM's evaluation code, ours (E01 records, mean $\pm$ s.d. over datasets) against the numbers published in CausalFM's Table 6 (which also reports the DR-learner at 1.485). Right: on B1 (F01 records), the median slope of the OLS fit of the estimated on the true CATE, the median ratio of estimated to true ATE, and mean $\sqrt{\text{PEHE}}$.

method	ACIC-10 ours	published	B1 slope	B1 ATE ratio	B1 $\sqrt{\text{PEHE}}$
CausalPFN	0.182 ± 0.08	0.239	1.00	1.00	0.579
CausalFM	0.619 ± 0.28	0.638	0.18	0.45	5.860
Do-PFN	0.789 ± 0.29	0.857	0.09	0.80	6.065
BCF	0.141 ± 0.08	–	0.59	1.00	2.666
DR-learner	0.156 ± 0.08	–	0.83	1.00	2.006

Table 28. Usage check R-E, continued. Top: mean $\sqrt{\text{PEHE}}$ (median CATE slope) on development cells that change only the effect (ATE level, effect s.d.) or the dimension (20 seeds each; Do-PFN v1 was pretrained on at most 6 features). Bottom: CausalFM under four input preprocessings, $\sqrt{\text{PEHE}}$ (median CATE slope; median ratio of estimated to true ATE) with the columns read as: ours (protocol; X and y standardized on observed training data), the toolkit's (scaler on all rows of X and on $y_0 \cup y_1$, which uses counterfactual outcomes), the toolkit's X scaler with our y scaler (B1 only), and raw X.

cell	CausalPFN	CausalFM	Do-PFN	BCF
ATE 0	0.214 (0.90)	0.396 (0.30)	0.444 (0.14)	0.200 (0.73)
base (ATE 0.5, s.d. 0.5)	0.214 (0.90)	0.368 (0.33)	0.445 (0.16)	0.198 (0.73)
ATE 2	0.214 (0.90)	0.435 (0.26)	0.601 (0.26)	0.194 (0.81)
ATE 4	0.214 (0.90)	0.750 (0.22)	0.997 (0.38)	0.198 (0.84)
effect s.d. 0	0.167	0.135	0.136	0.078
effect s.d. 1	0.245 (0.98)	0.698 (0.32)	0.854 (0.17)	0.284 (0.83)
effect s.d. 2	0.270 (0.99)	1.425 (0.31)	1.695 (0.13)	0.401 (0.91)
$d = 5$	0.188 (0.92)	0.312 (0.56)	0.394 (0.23)	0.183 (0.77)
CausalFM preprocessing	ours	toolkit ($y_0 \cup y_1$)	toolkit X only	raw X
CausalFM, base	0.368 (0.33; 1.14)	–	–	0.367 (0.34; 1.15)
CausalFM, B1	5.860 (0.18; 0.45)	4.582 (0.28; 1.01)	5.855 (0.18; 0.45)	6.884 (0.10; 0.21)
CausalFM, ACIC-10	3.313 (0.04; 0.65)	3.099 (0.04; 0.85)	–	5.285 (0.00; -0.12)

Table 29. Post-verdict analysis R-C (exploratory): repeated cross-fitting. Agreement of the deferral decision between two independent sets of splits (one split each; 5 each; 10 each), and, for the first set, mean $\sqrt{\text{PEHE}}$ (harmful share of deferrals; AUROC of the aggregated Z_{DR} for the failure event) with one split, with 10 splits aggregated by the median rule, and with 10 splits under the R-loss score. Deployed predictions are the same in every column. Two sets of 10 splits exist only at $n = 500$ (20 splits); at $n = 2000$ and on the B2 subset there are 10.

set	PFN	agree, 1	agree, 5	agree, 10	1 split	10 splits	10 splits, R-loss
H, $n = 500$	CausalPFN	83%	92%	92%	2.273 (13%; 0.844)	2.302 (8%; 0.860)	2.362 (0%; 0.908)
H, $n = 500$	CausalFM	83%	90%	96%	3.033 (10%; 0.827)	3.268 (9%; 0.852)	2.838 (7%; 0.898)
H, $n = 2000$	CausalPFN	91%	97%	–	1.529 (2%; 0.985)	1.585 (0%; 0.987)	1.603 (2%; 0.993)
H, $n = 2000$	CausalFM	89%	92%	–	2.091 (14%; 0.859)	2.097 (15%; 0.843)	2.003 (15%; 0.878)
H (all)	CausalPFN	87%	95%	–	1.901 (6%; 0.944)	1.943 (2%; 0.954)	1.983 (2%; 0.967)
H (all)	CausalFM	86%	91%	–	2.562 (13%; 0.859)	2.683 (13%; 0.864)	2.421 (13%; 0.898)
B2 subset (77)	CausalPFN	92%	99%	–	1.064 (15%; 0.836)	1.090 (18%; 0.826)	1.104 (11%; 0.899)
B2 subset (77)	CausalFM	100%	100%	–	1.097 (3%; 0.918)	1.085 (3%; 0.914)	1.323 (3%; 0.997)

Table 30. Post-verdict analysis R-D (exploratory): TabPFN-DR as a reference, cross-fitted on the same folds. Mean $\sqrt{\text{PEHE}}$ of always-PFN, always-TabPFN-DR, A-V5, A-V5 with $\mathcal{R} = \{\text{DR, BCF, TabPFN-DR}\}$ and with $\mathcal{R} = \{\text{TabPFN-DR}\}$ (share of tasks deferred to TabPFN-DR in some region; harmful share of deferrals).

bench	PFN	always-PFN	TabPFN-DR	A-V5	+TabPFN-DR	TabPFN-DR only
B1	CausalPFN	0.579	0.919	0.579	0.579 (1%; 0%)	0.581 (3%; 67%)
B1	CausalFM	5.860	0.919	2.105	1.097 (95%; 0%)	1.072 (100%; 0%)
B4	CausalPFN	14,383	13,757	14,329	13,998 (51%; 16%)	13,957 (56%; 16%)
B4	CausalFM	22,436	13,757	14,810	13,985 (96%; 0%)	13,945 (98%; 0%)
H	CausalPFN	2.130	1.318	1.901	1.800 (26%; 8%)	1.761 (43%; 7%)
H	CausalFM	3.839	1.318	2.562	2.045 (46%; 11%)	2.055 (68%; 12%)
B2 subset (77)	CausalPFN	1.307	0.900	1.064	0.924 (64%; 12%)	0.913 (78%; 12%)
B2 subset (77)	CausalFM	4.250	0.900	1.097	0.905 (83%; 4%)	0.905 (99%; 1%)

Table 31. Post-verdict analysis R2-A (exploratory): bias and noise of the paired statistics, over the tests whose null holds (oracle excess ≤ 0; for OW-DR and R-loss the overlap-weighted excess with the true e). With the truth, each estimate minus the oracle excess splits exactly into a bias B (conditional mean given X and the out-of-fold nuisances, minus the target) and a noise N. Rejected: by the actual test; with B removed; with N removed (Holm over the four tests in each case). Median B in units of the test's SE; for DR also with the true propensity clipped at 0.01 in place of $\hat{e}$ (B_{trunc}, the part due to truncation where Assumption A1 fails); share of tests with $B > 0$, which favours deferral. H without H5 / H6 and the platform without hidden-confounding cells, where $\mathbb{E}[Y \mid X, T] \neq \mu_T(X)$.

set	PFN	score	test	nulls	rejected	B removed	N removed	med B/SE	med B_{trunc}/SE	$B > 0$
B2	CausalPFN	DR	G_{out}, DR-l.	249	109	2	98	+2.95	+2.66	82%
B2	CausalPFN	DR	G_{out}, BCF	179	14	0	13	−0.17	0.00	45%
B2	CausalPFN	OW-DR	G_{out}, DR-l.	227	92	3	84	+1.06	–	74%
B2	CausalPFN	R-loss	G_{out}, DR-l.	227	31	4	26	+0.14	–	56%
B2	CausalFM	DR	G_{out}, DR-l.	38	28	2	25	+8.37	+8.29	89%
B2	CausalFM	R-loss	G_{out}, DR-l.	52	19	1	16	+0.65	–	71%
H (H1–H4)	CausalPFN	DR	all four	206	3	5	2	−0.16	0.00	32%
H (H1–H4)	CausalFM	DR	all four	88	2	3	0	−0.07	0.00	47%
platform	CausalPFN	DR	all four	1042	8	10	2	−0.06	0.00	45%
platform	CausalFM	DR	all four	450	33	6	13	+0.10	0.00	60%

Table 32. Post-verdict analysis R2-B (exploratory): the true-null classification of R-B with a margin. A test counts as a true null at margin k when its oracle excess is at most $-k$ times its estimated SE. Cells: rejected true nulls / true nulls (the same for the G_{out} test against the DR-learner; harmful deferrals that follow such a rejection). Last column: all harmful deferrals. R-loss and OW-DR use the overlap-weighted oracle.

set	PFN	score	$k = 0$	$k = 0.5$	$k = 1$	$k = 2$	harmful
B2	CausalPFN	registered	138/927 (109/249; 80)	114/697 (94/195; 72)	100/513 (82/148; 66)	76/283 (67/101; 54)	97
B2	CausalPFN	R-loss score	46/893 (31/227; 16)	8/405 (5/36; 2)	4/261 (1/17; 1)	1/104 (1/9; 0)	41
B2	CausalPFN	overlap-weighted DR	123/893 (92/227; 51)	15/390 (6/26; 6)	8/243 (3/15; 3)	2/95 (1/4; 0)	91
B2	CausalPFN	test G_{in} only	20/499 (0/0; 14)	11/370 (0/0; 7)	9/259 (0/0; 5)	2/109 (0/0; 1)	27
B2	CausalFM	registered	31/48 (28/38; 9)	23/30 (22/25; 8)	21/25 (21/22; 7)	20/22 (20/20; 7)	11
H	CausalPFN	registered	7/400 (1/126; 4)	4/237 (0/32; 4)	2/152 (0/5; 2)	1/64 (0/0; 1)	4
H	CausalFM	registered	23/275 (2/94; 19)	19/167 (0/31; 17)	18/108 (0/12; 17)	16/69 (0/1; 15)	19
platform	CausalPFN	registered	8/1145 (5/345; 5)	8/709 (5/135; 5)	5/487 (4/55; 3)	2/182 (2/17; 1)	6
platform	CausalFM	registered	38/599 (24/217; 19)	32/358 (22/81; 17)	22/247 (14/26; 14)	10/139 (4/9; 8)	20

I. Confirmatory Run of A-V5-R on ACIC-2017

The second review asked for one recommended variant, confirmed on new data. The variant is A-V5-R: A-V5 with the R-loss score of R-A in place of Equation (3), with the regions, references, folds, Holm family and choice rule unchanged. It was chosen after the confirmatory verdicts, from R-A, R2-A and the development data, which is why it needs this run. The protocol was pushed on 2026-10-14 at 21:16 UTC, before any method ran on these data.

Data. ACIC-2017 (Hahn et al., 2019), generated with the official `aiccomp2017` package: all 32 settings $\times$ replications 1–5, giving 160 tasks. Each task has the 58 covariates given to contestants and $n = 4302$ units, split 90/10 as for B2. The truth is the noiseless CATE returned by the generator. We recompute μ_0, μ_1 and the true propensity from the generator's formulas and draws, and check that they reproduce its output exactly. Only data-validity checks were run before the protocol was frozen. In 80 tasks some true propensities lie outside $[0.01, 0.99]$, on average 14% of the units, so Assumption A1 fails on half of the tasks, the regime of the B2 failure. ACIC-2017 shares its covariates with ACIC-2016 but not its outcome and treatment models.

Criterion (frozen). For each PFN, A-V5-R must satisfy two conditions: **C-1**, the one-sided 95% upper bound of its mean $\sqrt{\text{PEHE}}$ relative to always-PFN is at most 1.02; and **C-2**, at most 10% of the tasks on which it defers are harmed. The run confirms A-V5-R if both hold for both PFNs. The registered A-V5, the naive argmin, the always-reference baselines, TabPFN-DR (on one replication per setting) and the analyses R-B, R2-A and R2-B are reported descriptively.

Result. All 992 job records are complete and come from one commit (786cc4f). Before the analysis, the run's summary file was rebuilt from the saved job records with the harness's collector, because the driver's last call had overwritten it; the analysis code is the frozen one. A-V5-R passes both conditions for both PFNs, so F02 confirms it (Table 33). For CausalPFN the upper bound of the error ratio is 0.768 and 1/129 deferrals did harm; for CausalFM the figures are 0.401 and 0/122. The gain over always-PFN is significant for both after Holm (difference in mean $\sqrt{\text{PEHE}}$ -0.18, 95% CI $[-0.21, -0.15]$, for CausalPFN; -0.66, $[-0.76, -0.57]$, for CausalFM). The full-context output inside the wrapper equals the plain PFN (largest difference 7×10^{-9} for CausalPFN, 0 for CausalFM).

What F02 does not show. The case for A-V5-R rests on the size failure of A-V5 on B2 (R-B, R2-A), and that failure does not recur here. A-V5 rejects 0/110 true nulls for CausalPFN and 1/100 for CausalFM (A-V5-R: 0/104 and 0/94), does no harm (0/129 and 0/121) and is more accurate: 0.427 against 0.489 for CausalPFN, and 0.373 against 0.388 for CausalFM. The gap lies in the confounded settings, where CausalPFN's error ratio is 0.638 under A-V5 and 0.775 under A-V5-R; without confounding the two agree (0.648 and 0.650). This is consistent with the overlap weights of the R-loss, which give little weight to the units where overlap fails. In the R2-A decomposition, the DR score's median bias on the 47 true nulls in G_{out} against the DR-learner is -0.15 SE for CausalPFN (B2: +2.95), and its clipping part is -0.01 SE. Overlap fails on half of these tasks, so failed overlap alone does not produce the bias; in the algebra of R2-A, the clipping bias also scales with the outcome-model error at the clipped units. F02 therefore confirms that A-V5-R does no harm on new data, but it cannot tell A-V5-R from A-V5.

Deferring is also easy on these data. Both PFNs are much less accurate than BCF (always-PFN 0.666 and 1.052, always-BCF 0.341, always-DR-learner 0.561), and A-V5-R defers on 81% and 76% of the tasks. Even the naive argmin harms only 3/156 and 5/142 of its deferrals and comes close to always-BCF (0.341 and 0.343). Here the test's caution costs accuracy, not safety. On B1, where the PFN is accurate, the same naive argmin has $1.226\times$ the PFN's error. On the 32 tasks with one replication per setting, TabPFN-DR reaches 0.570, against 0.523 for A-V5-R with CausalPFN, 0.447 with CausalFM, and 0.678 and 1.045 for always-PFN. Under the R-loss, the full-sample statistic separates the protocol's failure events (104 and 118 tasks) with AUROC 0.898 and 0.983.

Table 33. F02, the confirmatory run of A-V5-R on ACIC-2017 (32 settings $\times$ 5 replications, 160 tasks; protocol frozen before any method ran on these data). Mean $\sqrt{\text{PEHE}}$, its ratio to always-PFN, the one-sided 95% upper bound of the ratio (C-1: ≤ 1.02), the share of tasks with a deferral, and harmful deferrals (C-2: $\leq 10\%$). The registered A-V5 and the naive argmin are descriptive.

PFN	method	mean $\sqrt{\text{PEHE}}$	ratio	NI upper	deferred	harmful
CausalPFN	A-V5-R (tested)	0.489	0.734	0.768	81%	1/129 (0.8%)
CausalPFN	A-V5 (registered)	0.427	0.641	0.673	81%	0/129 (0.0%)
CausalPFN	naive argmin, R-loss	0.341	0.513	0.536	98%	3/156 (1.9%)
CausalPFN	always-PFN / DR-learner / BCF	0.666 / 0.561 / 0.341				
CausalFM	A-V5-R (tested)	0.388	0.369	0.401	76%	0/122 (0.0%)
CausalFM	A-V5 (registered)	0.373	0.354	0.382	76%	0/121 (0.0%)
CausalFM	naive argmin, R-loss	0.343	0.326	0.349	89%	5/142 (3.5%)
CausalFM	always-PFN / DR-learner / BCF	1.052 / 0.561 / 0.341				

J. Reproduction of Published Numbers

Table 34. Reproduction under the CausalPFN paper's protocol: $\sqrt{\text{PEHE}}$ (mean ± s.e.; IHDP and LaLonde 100 realizations, ACIC 10) with published values where reported. ACIC-2016 is the 10-dataset causallib copy; LaLonde values are ITE-RMSE against noisy ITEs. src: `runs/e01_repro_cate/analysis/repro.md`.

method	ihdp	acic2016	lalonde_cps	lalonde_psid
BCF	2.67±0.44	0.739±0.12	1.28e+04±67	2.26e+04±1.5e+02
CausalFM	5.86±0.73	3.31±0.53	1.24e+04±16	2.24e+04±1.3e+02
CausalPFN	0.579±0.07 / pub 0.58±0.07	0.962±0.13 / pub 0.92±0.11	8.95e+03±21 / pub 8960±20	1.44e+04±2e+02 / pub 14400±200
CausalPFN-cal	0.579±0.07	0.962±0.13	8.86e+03±54 (n=6)	—
CF	3.8±0.62	1.17±0.25	1.34e+06±4e+04	4.83e+04±2.2e+03
Do-PFN	6.07±0.9	4.25±0.56	1.2e+04±32	2.09e+04±1.4e+02
DR	2.01±0.27	0.823±0.13	9.17e+03±26	1.49e+04±2.8e+02
TabPFN-T	0.793±0.15 / pub 0.95±0.2	0.498±0.085 / pub 0.54±0.08	— / pub 9450±190	— / pub 18700±830

Table 35. Relative ATE error under the CausalPFN paper's protocol with the correct per-realization truth; the last column scores IHDP against the constant 4 used by the published loader. src: `runs/e01_repro_cate/analysis/repro.md`.

method	ihdp	acic2016	lalonde_cps	lalonde_psid	IHDP rel@4
BCF	0.0513±0.021	0.0306±0.0083	0.99±0.014	0.999±0.0086	0.197±0.042
CausalFM	0.596±0.083	0.338±0.036	0.942±0.00019	0.95±0.0011	0.5±0.019
CausalPFN	0.0233±0.0027 / pub 0.2±0.04	0.0506±0.012 / pub 0.05±0.01	0.132±0.0099 / pub 0.13±0.01	0.21±0.015 / pub 0.22±0.02	0.203±0.038
CausalPFN-cal	—	—	—	—	—
CF	0.111±0.025	0.0303±0.014	14.9±1.3	1.25±0.059	0.109±0.018
Do-PFN	0.51±0.14	0.696±0.042	0.874±0.0061	0.916±0.006	0.572±0.096
DR	0.0311±0.0058	0.0305±0.014	0.22±0.013	0.0762±0.0079	0.203±0.039
TabPFN-T	0.0245±0.0027 / pub 0.21±0.04	0.0139±0.0041 / pub 0.03±0.01	— / pub 0.32±0.05	— / pub 0.6±0.07	0.211±0.039

On IHDP, CausalPFN's published loader assigns the ATE 4 to all 100 realizations; the per-realization truth (mean $\mu_1 - \mu_0$ over training and test units) ranges from -0.60 to 12.65 with median 4.09. Scored against the constant, our harness reproduces the published relative ATE error of CausalPFN (0.203 ± 0.038 vs. 0.20 ± 0.04) and of TabPFN-T (0.211 vs. 0.21). On CausalFM's own evaluation data, CausalFM reaches 0.479 ± 0.050 on its ten synthetic datasets against CausalPFN 0.454, BCF 0.456 and causal forest 0.458; on Jobs our numbers are within 4% of the published ones. CausalFM's preprocessing fits the outcome scaler on both potential outcomes, a counterfactual leak; Appendix H (R-E) shows that it is not small on IHDP. Under fixed configurations the classical baselines can fail silently: on LaLonde-CPS (154 treated of 14.4k units) the causal forest's ITE-RMSE explodes (median 1.3×10^6 vs. $\approx 9 \times 10^3$ for the other methods) and BCF shrinks the effect to zero (median |ATE err| 7037 for a true ATE of about -7041).

K. Compute and Software

Code for the harness, the method, the held-out suite, the analysis scripts and the frozen protocol is provided as anonymous supplementary material and will be released with the camera-ready version. All runs used CPUs only (Intel Xeon Platinum 8555C, 4 threads per job, at most 48 concurrent jobs on a shared server); no GPU was used at any stage. Software: Python 3.11, `causalpfn` 0.1.5 with the released weights, Do-PFN at its released commit (separate environment with PyTorch 2.1), the CausalFM toolkit and checkpoint at their released commit, `tabpfn` 2.0.9 with v2 weights, EconML 0.17, LightGBM 4.7, stochtree, the authors' MP-OSPC code, and R 4.4.3 with the official `aciccomp2016` and `aciccomp2017` generators (vdorie/aciccomp at 282d266). Every job record stores the code version, configuration, seed, thread count, wall time and peak memory, and every number in this paper is regenerated from these records by the released analysis scripts.